

Hours-Biased Technological Change*

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September 2026

Abstract

In the United States, average hours worked have declined over the long run, but unevenly across the wage distribution: hours have fallen at the bottom and risen at the top, reversing the hours-wage correlation. Motivated by these facts, this paper proposes a matching model of the labor market in which workers choose hours and sort into jobs that differ in how they reward working time. Hours-Biased Technological Change (HBTC) increases the complementarity between hours and the worker-job bundle, so the return to hours increasingly operates through the allocation of workers to jobs. HBTC can overturn income effects when high-skill workers are allocated to jobs where their hours are especially productive. An estimated version of the model using U.S. data over the post-1980 period reveals that HBTC accounts for about a third of the model-implied rise in income inequality and explains the rise in the hours-wage correlation.

Keywords: Hours; Sorting; Technological Change; Income Inequality.

JEL codes: J22, J31, C78, O33.

*I thank my advisors: Isaac Baley, Jan Eeckhout, Edouard Schaal, for discussions, guidance and support. I thank Julieta Caunedo, Hector Chade, Ana Figueiredo, Chris Edmond, Christoph Hedtrich, Josep Pijoan-Mas, Maria Ptashkina, Michelle Rendall, Alireza Sepahsalari, Lawrence Uren, Gianluca Violante, David Zentler-Munro for detailed feedback on the paper, and various seminar audiences for helpful comments. All errors are my own.

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1 Introduction

Over the last century, hours worked have declined substantially. For example, the average American worker now spends 26% less time at work than she did a century ago. Given that sustained economic growth has increased productivity and wages, this suggests that workers devote more time to leisure as they grow richer (Boppart and Krusell, 2020).¹

However, recent cross-sectional evidence suggests that *who* spends more time at work changes as countries develop. In the U.S. of the 1950s, for example, those with the lowest incomes worked the longest hours. Nowadays, this correlation has reversed: it is CEOs, top lawyers, and other high-wage workers who work the most. This observation is hard to explain in light of the long-run evidence: If higher wages induce to work less, why is it the case that higher-wage workers work more, and increasingly so?

This paper proposes a new mechanism to reconcile these facts, and explores its implications. The theory builds on the observation that the systematic increases in the hours-wage correlation have been accompanied by unprecedented technological progress. It has long been recognized that technological change has been behind changes in the wage structure over the past decades, with computers and ICTs as prominent examples. But new technologies have also changed what an hour of work can do: they made work less tied to a particular workplace or a fixed schedule, and more easily continued across workers, ultimately making it easier to adjust the work schedule with extra hours if required. For example, a consultant in New York can now use shared cloud files or instant communication to coordinate early in the morning with Europe and stay online late to resolve issues with Asia, turning what used to be separate business days into one continuous production window.² These changes are pervasive in modern labor markets, but are unlikely to affect all workers equally: a late-night online meeting between the top managers of a ride-sharing company via smartphone brings more value to the firm than an additional ride from one of the drivers.³

¹The fact that the rich used to consume more leisure has attracted the attention of sociologists since at least the end of the 19th century: Veblen (1899) discusses how the wealthiest were spending large parts of their time in leisure activities, while low-class people worked long hours.

²Organizational research has studied the “autonomy paradox”, a phenomenon by which the introduction of mobile devices has offered flexibility on one hand, and at the same time, increased availability expectations, causing high-skilled professional workers to be effectively less flexible in their working time (Mazmanian *et al.*, 2013). Presser (2003) provides an early account of how economic activity increasingly moved into evenings, nights, weekends, and across time zones as service demand expanded and communication costs fell.

³While most evident across workers at opposite ends of organizations, the same logic can apply even within skilled work. Shared patient records and clinical workflows can make some skilled tasks easier to transfer while extending the reach of senior decision-makers: residents and pharmacists can pass routine steps across shifts, while the same system lets an attending physician see the full history remotely and intervene when a complex case requires a decision.

This project provides a framework to study how technological change affects the returns to working hours across workers and jobs – and the associated implications for inequality and the macroeconomy.

The first contribution is theoretical. The theory is a matching model of the labor market in which workers sort into jobs based on skill as well as how their time is valued in that job.⁴ In the model, an income effect induces high-skilled workers to work fewer hours; at the same time, complementarities in production raise the returns to hours worked in more productive jobs. When high-skilled workers are allocated to those jobs, this implies that their returns to hours are higher, and the more so the stronger the complementarities between hours and job/skill attributes. When such complementarities are strong enough, high-skilled workers supply more hours despite the large income effects. The same trade-off between income effects and hours complementarities determines the equilibrium sorting condition: when better jobs reward hours more, the hours choice of high-skilled workers reinforces the direct type-complementarities, and can overcome income effects that push in the opposite direction. As part of the theoretical contribution, I characterize the equilibrium link between the skill-job allocation and hours choice for a wide class of preferences and production functions. Furthermore, I show that the framework nests several models typically used in the literature on labor supply, and highlight that these models typically fail to account for a positive and rising hours-wage correlation.

Technological change affecting working hours (a phenomenon we label Hours-Biased Technological Change, HBTC) is what magnifies the link between hours and sorting. In the model, a change that increases the complementarity between hours and the skill-job bundle raises the value of an additional hour especially in better jobs. With positive assortative matching, stronger hours complementarities make high-skilled workers choose longer hours and increase their earnings by more, because they are allocated to the jobs where hours are most valuable. In this sense, the model's key insight is that the true returns to hours reflect not only a within-job return, but also include the effect that working more has on the allocation of workers to jobs. This channel is especially strong when new technologies make jobs increasingly differ in how they reward working time.

As a second contribution, the paper uses U.S. data over the post-1979 period to estimate the model. Estimated parameter changes reveal that well-known forces, such as increases in marginal returns to skill (i.e. Skill-Biased Technological Change, [Katz and Murphy \(1992\)](#), [Krusell, Ohanian, Ríos-Rull and Violante \(2000\)](#), [Autor and Acemoglu \(2011\)](#)), have been accompanied by a change in the relative value of hours worked across

⁴For a framework featuring endogenous hours with sorting, see [Calvo, Lindenlaub and Reynoso \(2024\)](#) and [Michelacci and Pijoan-Mas \(2016\)](#).

skills and jobs, which is interpreted as HBTC.⁵ Thus, the post-1980 technology changes were not only skill-biased, but also hours-biased, in the sense that the value of an additional hour increasingly depended on the worker-job match in which it was supplied.

The estimated model is used to decompose the rise in income inequality. SBTC and HBTC both increase inequality, but through different channels. SBTC is the main force behind the rise in wage dispersion. HBTC is a new source of inequality because it raises the covariance between wages and hours: high wages and long hours increasingly occur together. This channel is essential because the large income effects embedded in preferences push in the opposite direction. Without HBTC, SBTC would raise wages at the top but induce high-wage workers to reduce hours, causing the wage-hours covariance to fall. Conversely, without SBTC, the covariance effect of HBTC would be larger, but the rise in wage dispersion would be much smaller.⁶ The estimates also show that the quantitative force of HBTC comes through sorting: When the dispersion in job types and type-complementarities are held fixed at their 1979 levels, changing the return to hours has a much smaller effect. Once the estimated changes in type-complementarities and dispersion in job productivities are allowed to take place, high-skilled workers' additional hours are most productive, and the effect of HBTC on income inequality becomes much larger.

I provide suggestive evidence on the main mechanism. Building on [Denning, Jacob, Lefgren and vom Lehn \(2022\)](#), I replicate in my CPS-ORG sample their distinction between within-occupation returns to actual hours and across-occupation prices of expected hours. The results mirror theirs: within occupations, the wage-hours elasticity remains close to zero, while the across occupation elasticity is large and rises over time. This pattern points to sorting as an increasingly important source of the return to hours. In particular, the across-occupation elasticity captures the wage differences associated with workers' allocation to occupations with different hours requirements, rather than the return to varying hours within the same occupation. This evidence supports the mechanism because it shows that the return to an additional hour includes not only its payoff within a given job, but also its effect on the set of jobs to which the worker can be allocated. Importantly, this margin has become stronger over time.

To conclude, two further exercises illustrate broader implications of the mechanism.

⁵In particular, estimates reveal that in 1980, hours and skills/jobs were roughly Cobb-Douglas in production. Starting from the 1980's, and as a consequence of HBTC, these complementarities have become stronger and estimates suggest that hours and skills/jobs are complements.

⁶These effects are quantitatively significant: In counterfactual exercises, HBTC is shown to account for about 34% of the model-implied increase in income inequality, thus confirming the importance of correctly modeling hours in production to fully understand the drivers of income inequality in U.S.

First, I extend the model with preference shocks, extending Galichon, Kominers and Weber (2019) and Bandiera, Kotia, Lindenlaub, Moser and Prat (2024), to study constraints on long hours when job assignment is stochastic. When HBTC makes hours especially productive in top matches, these constraints can move high-skill workers away from the jobs where their hours are most valuable. Second, I compare fixed-assignment earnings elasticities with full-assignment elasticities that allow hours to affect job allocation, in the spirit of Scheuer and Werning (2017). The main insight is that HBTC changes the earnings elasticity relevant for progressivity calculations: when hours affect access to better jobs, the response to incentives reflects both hours choices and movement along the assignment schedule, with HBTC increasing the average proportional adjustment at the top of the earnings distribution.

Literature A vast literature studies the relationship between hours and wages across countries and time periods. Boppart and Krusell (2020) introduce a new general preference class in which income effects dominate substitution effects. Relatedly, Bick, Fuchs-Schündeln and Lagakos (2018) and Bick, Fuchs-Schündeln, Lagakos and Tsujiyama (2022b) highlight the role of structural change in accounting for patterns of hours worked along the development spectrum. Rachel (2021) and Kopytov, Roussanov and Taschereau-Dumouchel (2023) propose theories to explain the decline in hours worked based on improvements in leisure technologies.⁷ Relative to all these papers, this study argues that technological change can systematically change the returns to hours across jobs as countries develop, and reconciles the aggregate and cross-sectional evidence by complementing preference-based theories with a theory in which technological complementarities play a central role.

This paper also closely relates to the literature that has attempted to explain diverging trends in labor market outcomes between high- and low-skilled. Wolcott (2021) proposes a model to explain the diverging trends in employment rates across skills. Boppart and Ngai (2021) reconcile diverging trends in leisure and increasing inequality with a mechanism based on intertemporal substitution. Heathcote, Storesletten and Violante (2010) analyze the macroeconomic and welfare implications of rising wage inequality and increasing hours-wage correlation. Auray, Fuller and Vandenbroucke (2021) study multiple jobs and comparative advantage in a model that embeds a similar tension between cross-sectional and time series facts on hours worked, as in this paper. Alon, Coskun and Doepke (2019) propose a family channel to explain the reversal in the hours-wage correlation. Ferraro and Valaitis (2025) reconcile the flat cross-sectional employment rates with

⁷Hochmuth (2024) focuses on the movements in hours worked among entrepreneurs.

a model of quality choice. Relative to these papers, I contribute a new mechanism and show its quantitative importance in explaining rising inequality.

With respect to models of sorting (Lindenlaub, 2017; Chade and Lindenlaub, 2022; Eeckhout and Kircher, 2018; Vereshchagina, 2021), this paper explicitly models the hours decision. In this respect, the following contributions are closest to this framework: Calvo, Lindenlaub and Reynoso (2024), Michelacci and Pijoan-Mas (2016) and Shao, Sohail and Yurdagul (2026). The first studies how the interplay between marriage and labor market decisions, shaped by the endogenous hours decision, affects inequality and hours in equilibrium. The second provides a competitive growth model where the pace of technological progress affects hours and the sorting of workers to jobs. The third studies the role of hours complementarities in production on the size distribution of firms, and their impact on earnings inequality. Relative to these papers, this paper characterizes sorting for a general class of production and utility functions and proposes a new mechanism to reconcile the data.⁸

2 Motivating Evidence

In this section, I present two sets of facts (one aggregate and one cross-sectional) that motivate the theory. Both facts concern weekly hours worked and hourly wages; for the sake of brevity, detailed variable definitions are listed in Appendix A.1.

Data The main data source I use for hours worked in U.S. is the CPS, which contains detailed information on hours worked and wages for the period of interest. I complement hours data from CPS with hours data from Kendrick (1961) and Kendrick (1973) to be able to trace total hours worked back in time⁹. In Appendix A, I show that the main messages delivered by the analysis below are very similar across other datasets of hours worked (e.g. ATUS).¹⁰ Data on hours worked across countries are from Penn World Tables.

⁸Moreover, I differ in the quantitative application of the paper, in that I estimate how the production technology has changed over time, with a particular focus on how hours enter in production. In this sense, another related paper is Erosa, Fuster, Kambourov and Rogerson (2022), which develops a Roy-style model of occupational choice to study the implications of gender differences in home production possibilities for the gender wage gap.

⁹The latter datasets have been used extensively in the labor supply literature to capture long-run trends in hours worked (see e.g. (Ramey and Francis, 2009) and (Cociuba *et al.*, 2018)).

¹⁰To construct a real wage index, I use labor productivity divided by hours worked and complement it with series from Kendrick (1973). The same approach is used by Kopytov *et al.* (2023).

Average Hours Worked The first motivating fact highlights the long-run behavior of hours per worker.¹¹ Figure 1 plots, for the U.S. and for selected advanced economies, the long-run behavior of hours worked per worker, together with an index of real average wages for the U.S. Over the long-run, average hours worked exhibit a clear downward trend across countries. For the U.S., the downward trend in hours is accompanied by a steady rise in average wages (left panel). The magnitudes are also significant, with some countries observing declines in average hours of roughly ten hours (see e.g. Denmark, Germany, U.S.).

These simple patterns, interpreted through the lens of neoclassical models of labor supply, point towards preferences where income effects are stronger than substitution effects, so that an increase in average wages due to productivity implies lower average hours worked. This is the main intuition behind models such as Boppart and Krusell (2020) and Bick *et al.* (2022b). Recent evidence on hours trends along the development spectrum is also consistent with this idea: Bick *et al.* (2018) show that hours per worker decline with GDP per capita, and that these patterns cannot be explained by systematic differences in age, educational attainment or sectoral composition across countries.¹²

Hours and Wages in the Cross-Section The second set of facts concerns the cross-sectional relationship between hours worked and wages. Using CPS-ORG data, I divide the wage distribution into ten deciles and plot the mean wages and hours for three representative deciles¹³ (variables definition and sample selection details are listed in Appendix A.1). Figure 2 clearly shows diverging patterns for hours and wages across the distribution: while wages for high-wage workers rose, wages at the bottom stagnated, consistent with rising wage inequality (Autor and Acemoglu, 2011) driven by wages disproportionately increasing at the top. Hours worked also diverged across the distribution: high-wage workers started working longer hours, with an increase concentrated in the 1980's and the

¹¹Since the focus of the paper is on the intensive margin, the relevant measure to be considered is hours per worker. Moreover, as noted in Boppart and Krusell (2020), long run movements in weekly hours worked per worker (the intensive margin) are more important than participation rates (the extensive margin). Nevertheless, a similar picture emerges if we consider hours per capita over the very long term (see also Ramey and Francis (2009)).

¹²Moreover, these trends are consistent with papers using time use data, which show that hours worked have declined (and leisure time increased) in U.S., as in Aguiar and Hurst (2007). Similar evidence is also provided in Kopytov, Roussanov and Taschereau-Dumouchel (2023), which show that the overall decrease of hours as wages grow is a trend that characterizes all OECD countries in the post-war period.

¹³Wage deciles are computed separately in each year and therefore identify positions in the contemporaneous wage distribution, rather than fixed individuals or worker types. This is the relevant classification for the inequality decomposition, which depends on the contemporaneous covariance between wages and hours, but has the drawback that workers in the same decile across different years might not be comparable. For this reason, the analysis below complements this rank-based evidence with results for fixed education groups.

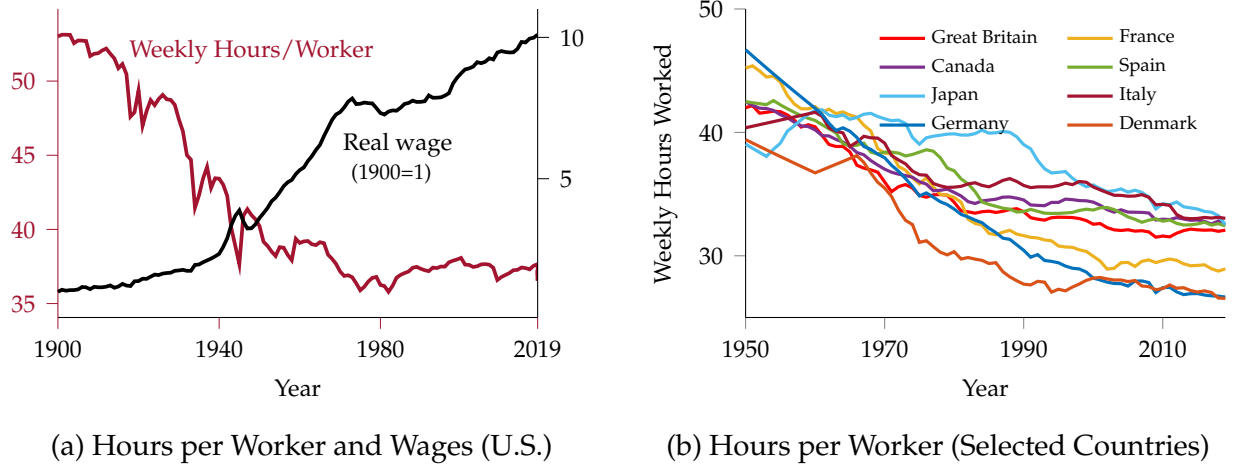


Figure 1: Working time trends in the United States and other advanced economies.

Notes: Panel (a) plots average weekly hours worked per worker in the United States together with a real wage index (normalized to 1 in 1900). For 1900-1958, total annual hours and total employment are from [Kendrick \(1961\)](#) and [Kendrick \(1973\)](#), taken from the [Ramey and Francis \(2009\)](#) replication files, with average weekly hours computed as total annual hours divided by total employment, divided by 52; for 1959-2019, total annual hours are from the CPS-based series in [Cociuba et al. \(2018\)](#), divided by the annual mean of monthly civilian employment from the BLS, divided by 52. The wage index is constructed following [Kopytov et al. \(2023\)](#), computed as real GDP (1900-1928) and real compensation of employees (1929-2019), divided by employment and by the hours-per-worker series. Panel (b) plots average weekly hours worked per employed worker in selected economies (1950-2019), constructed from average annual hours worked per person engaged in the Penn World Table ([Feenstra, Inklaar and Timmer, 2015](#)), divided by 52 (retrieved from FRED).

1990's; viceversa, low-wage workers saw their hours worked steadily decrease.¹⁴ Similar patterns can be observed when categorizing workers by their skill (proxied by education) rather than by wage deciles, as shown in Figure A.3 in Appendix A.2: regardless of their skill, workers decreased their market hours uniformly until 1985; after that point, hours worked diverged across education groups.

To formalize the qualitative results displayed in Figure 2, I analyze the cross-sectional hours-wage elasticity by running the following regression year-by-year in the baseline sample:

$$\log(h_i) = \alpha + \beta \log(w_i) + X_i + \epsilon, \quad (1)$$

where h_i is individual hours worked, w_i is individual wages, and X_i are demographic controls¹⁵. The estimated coefficient $\hat{\beta}_t$ summarizes the relationship between hours and

¹⁴These patterns are consistent with [Costa \(2000\)](#), who showed that the U.S. experience is characterized by a sign reversal in the hours-wage elasticity, from negative to positive: low-wage workers used to work the longest hours in 1960's and 1970's; starting from the 1980's, however, high-wage workers work significantly more hours than low-wage workers.

¹⁵In the baseline specification, I consider age and age squared as controls to account for the systematic variation in hours worked across the life-cycle. This is the same specification used in [Costa \(2000\)](#) but

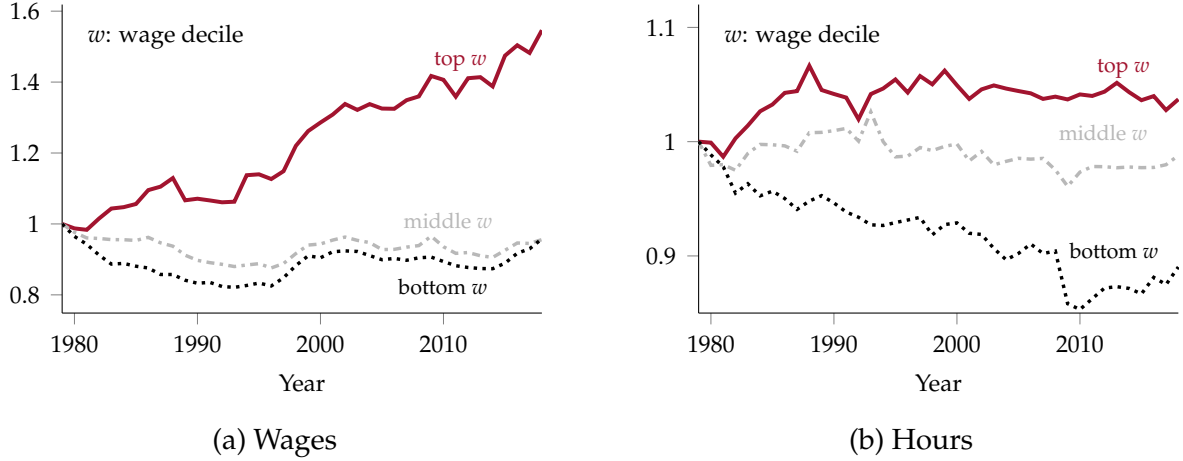


Figure 2: Wages and Hours by Wage Decile (US).

Notes: The left (right) panel displays mean wages (hours) by wage deciles. "Top w " (red continuous line) represents the mean outcome among workers in the highest-wage decile (the top 10 percent); "Middle w " (grey dashed line) reports the mean outcome among workers in the fifth wage decile (between the 40th and 50th percentiles); "Bottom w " (dotted black line) reports the mean outcome among workers in the lowest-wage decile (the bottom 10 percent). Series are normalized by dividing by their value at the beginning of the sample. Raw series are plotted in Figure A.2. Source: CPS-ORG and own calculations. For details, see Appendix A.1.

wages in the cross-section¹⁶. The estimated coefficient for the U.S. over time is depicted in Figure 3 (left panel). The right panel shows, using data from Bick *et al.* (2018), the coefficient obtained from the regression (1) performed across countries, and it is plotted against the level of per capita GDP, denoted by $\hat{\beta}_{cc}$.

For the U.S., the coefficient is negative at the beginning of the sample, and - starting from the 1980's - turns significantly positive and increasing, with a mild decrease only in the most recent years in the sample (left panel).¹⁷ This trend (rising, from negative to positive) is robust to hours definitions, sample selection, and is essentially unchanged when worker controls (education and race) as well as industry controls are added, as shown in the robustness checks contained in Appendix A.2.

As first shown in Bick *et al.* (2018), the U.S. experience is replicated across countries along the development spectrum: as the right panel shows, the hours-wage elasticity is

updated to more recent years, and the same specification adopted in Bick *et al.* (2018).

¹⁶As such, the coefficient should be interpreted as descriptive of the correlation between hours and wages, and does not represent a structural elasticity.

¹⁷Similar evidence is provided by Heathcote *et al.* (2010) and Boppart *et al.* (2024) using U.S. data. Moreover, data from time use sources (see in particular Aguiar and Hurst (2007)) are consistent with the evidence presented thus far: starting from the mid-1980's, leisure time has increased for the low-skilled workers more than for the high-skilled workers (while importantly, it increased uniformly across skill categories between 1965 and 1985). Figure A.3 in Appendix A.2 shows this explicitly by replicating the results in Boppart and Ngai (2021).

negative for low-income countries, while it gets progressively smaller and even positive for the richest countries. In other words, as technology evolves, it is high-wage workers that progressively work the longest hours, while low-wage workers work fewer hours (and increasingly so). Importantly, while we restrict the sample to males-only, these patterns hold across genders, as shown in [Bick *et al.* \(2018\)](#).

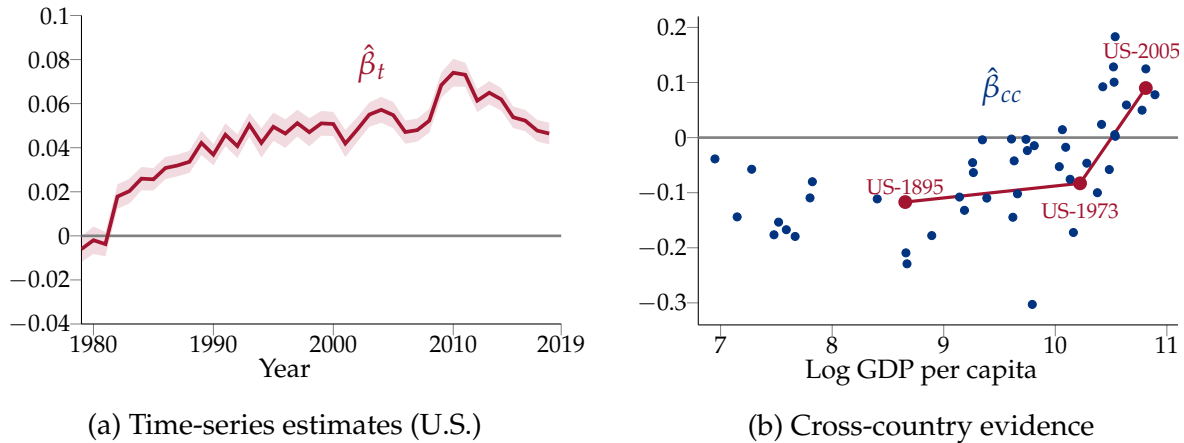


Figure 3: Elasticity of hours with respect to wages.

Notes: Panel (a) plots the estimated coefficient $\hat{\beta}_t$ from the regression of log hours on log wages over time run with US data (CPS-ORG), as in (1); the shaded area denotes the 95% confidence interval. Panel (b) plots the cross-country elasticity estimates from [Bick *et al.* \(2018\)](#) against log GDP per capita. The red line connects the historical U.S. estimates from [Costa \(2000\)](#) for 1895 and 1973 with the U.S. 2005 estimate from the [Bick *et al.* \(2018\)](#) dataset.

Summary of the Evidence Figure 2 and Figure 3 have shown that the negative relationship emerging from aggregate data (and suggestive of preferences where income effects exceed substitution effects) seems to be counteracted by other forces in the cross-section, and this is especially true for the U.S. in recent years and for countries at the technological frontier. This observation suggests a link between technological progress and the cross-sectional relationship between hours and wages. Importantly, the concurrent increase in wage inequality (especially in the U.S.) only makes these patterns harder to explain, since the forces that typically explain declining hours in the aggregate would predict large decreases in hours for high-wage workers.

Taken together, the evidence motivates a theory that builds on preference specifications as in [Boppart and Krusell \(2020\)](#) and at the same time attempts to reconcile the aggregate evidence with the evolving cross-sectional relationship between hours and wages.¹⁸ The next section develops a theory to investigate this relationship and how

¹⁸Models explaining wage inequality have shown how technological change can affect the marginal

it's affected by technological change.

3 Theory

This section develops a new assignment framework, in which workers sort into jobs based on their skills and hours.¹⁹ After characterizing the theory, I show how it nests several models of the labor supply as special cases. Finally, I do comparative statics to illustrate the mechanism.

3.1 Framework

Environment I consider a static competitive labor market, composed of heterogeneous workers $x \in \mathcal{X} = [0, \bar{x}]$ and jobs $y \in \mathcal{Y} = [0, \bar{y}]$; we denote by $Q(x)$ and $G(y)$ the distribution of workers and jobs, respectively, and $q(x)$ and $g(y)$ as their densities. Agents on both sides of the market have mass equal to one. The attribute of individuals x can be thought of as time-invariant skill or ability; the productivity of the job y can be thought of as a skill requirement of a particular job. Workers and jobs match in a one-to-one fashion to produce output, denoted by $f(x, y, h)$. Output f depends, therefore, on both the quality of jobs y and workers x assigned to them, and on the hours supplied by workers, h .

Individuals are endowed with one unit of time to be allocated between market work and leisure (there is no home production). Individuals get utility from consumption and leisure, and we denote utility by $u(c, h)$, which is allowed to be non-separable in its inputs.

Jobs' problem Jobs maximize profits by choosing the type x and hours h ; profits are defined as output net of income w to be paid to the worker²⁰:

$$\max_{x, h} f(x, y, h) - w \quad (2)$$

The choice of worker type x by the job will determine, in equilibrium, the assignment function $\mu(x)$ that maps workers to jobs as well as income and profit functions.

product of workers' skills (Krusell *et al.*, 2000) and the allocation of skills to jobs (Autor and Acemoglu, 2011).

¹⁹For the purposes of the theoretical section, one can view y as firms or jobs, interchangeably, as the theoretical model allows for both interpretations.

²⁰To save on notation, I use w as income but it should be understood that income is a function of the matched pair x, y and the hours choice, i.e. $w = w(x, y, h)$. In equilibrium, with endogenous assignment $\mu(x)$ and hours $h(x)$, income will be a function of workers' skills $w(x)$.

Workers' problem Workers choose time allocation taking income w as given:

$$\max_h u(c, h) \quad \text{s.t.} \quad c = w \quad (3)$$

Together with the jobs' problem, this determines optimal choice of hours h as a function of skill, $h^*(x)$. Hours choice then becomes the key link between the worker's problem and the jobs' problem. If output f only depends on skills x and job attributes y (that is, $f = f(x, y)$), this is a standard assignment problem between workers and jobs in the tradition of [Becker \(1973\)](#). Precisely because hours enter in production, the labor supply choice by the worker affects the equilibrium sorting pattern and with it, the equilibrium levels of inequality and output. Therefore, hours choice and equilibrium assignment are intertwined and jointly determined by the jobs' and workers' problems.

Market clearing The model is closed by specifying a market clearing condition, essentially requiring that the workers and jobs match in a measure-preserving way. Under Positive Assortative Matching (PAM), better workers are assigned to better jobs and the market clearing condition can be written as²¹:

$$\int_{\mu(x)}^{\bar{y}} g(s) ds = \int_x^{\bar{x}} q(s) ds \quad (4)$$

where $\mu(x)$ is the equilibrium matching function, assigning workers to jobs.

Equilibrium We are now ready to define a competitive equilibrium of this economy.

Definition 1. *A competitive equilibrium of this economy is a tuple of functions $(w(x), \mu(x), h(x))$ such that:*

- $w(x)$ and $h(x)$ solve problems (2) and (3) (optimality)
- equation (4) holds (market clearing)

3.2 Conditions for Assortative Matching

Towards a complete characterization of the equilibrium of this economy, we want to seek conditions under which assortative matching arises. This is most directly achieved by noticing that this is effectively a matching problem with imperfectly transferable utility

²¹Here and throughout the text, I refer to PAM as Positive Assortative Matching, indicating that better workers are matched in equilibrium with better jobs (analogously for Negative Assortative Matching, NAM).

(ITU) between matched agents. This class of problems has been analyzed in the literature (see in particular, [Legros and Newman \(2007\)](#) and [Chade *et al.* \(2017\)](#)).

To apply the results from the literature on matching with ITU, it is useful to rewrite the problem by substituting the wage in the worker problem (3), assuming that firm y gets at least a value $V \in [0, V^{max}]$ when matched with worker x . The pair's problem then becomes:

$$\Phi(x, y, V) = \max_h u(f(x, y, h) - V, h) = u(f(x, y, h^*) - V, h^*) \quad (5)$$

where $\Phi(x, y, \cdot)$ traces out the Pareto utility frontier of the match, for every possible value of V . Of course, in equilibrium, V will be a function of the firm's attribute ($V = V(y)$) and is determined endogenously with the equilibrium assignment.

The usefulness of writing the problem as in (5) is that in order to know whether PAM arises in equilibrium, it is enough to check that the Generalized Indifference Condition (GID) provided in [Legros and Newman \(2007\)](#) is satisfied. Because in this setting agents' types are continuous, we will use the differential version provided in [Chade *et al.* \(2017\)](#). In particular, PAM arises in equilibrium if (and only if):

$$\Phi_{xy} - \Phi_{xV} \frac{\Phi_y}{\Phi_V} > 0 \quad (6)$$

Equation (6) makes clear that two types of complementarities drive equilibrium sorting.²² There is a complementarity between types, Φ_{xy} : if better workers do disproportionately better with better jobs ($\Phi_{xy} > 0$), this is a force pushing towards PAM. In the case of transferable utility (TU), $\Phi_{xy} = f_{xy}$ and $\Phi_{xV} = 0$, therefore (6) boils down to $f_{xy} > 0$ (as in [Becker \(1973\)](#)). In this ITU setting, however, what matters for sorting is also how the surplus of the match varies across worker-firm pairs, which in turn depends on the complementarity between worker type and partner's (jobs') utility, captured by Φ_{xV} . Intuitively, this term measures how easy it is to transfer utility between higher x workers and higher y jobs. The easier it is to transfer utility (the larger Φ_{xV}), the more likely it is that (6) is satisfied (recall that $\Phi_V < 0$). In this setting, this force will depend on both characteristics of preferences (income effects) and on technological complementarities between hours, jobs and worker types. The next step is to highlight these characteristics by writing (6) in terms of primitives of the model (preferences and technology).²³ This leads

²²Equation (6) can be derived from the FOC of (5) with respect to job type y , $\Phi_y + \Phi_V V_y = 0$, and then differentiating this along the equilibrium matching $\mu(x)$ combined with the requirement that the SOC must be negative for this to be an optimum.

²³Alternatively, the problem can be characterized as a joint, simultaneous choice of jobs and hours by the workers, that gives rise to a multidimensional second order condition; then, PAM/NAM arises depending

us to the first proposition:

Proposition 1. *A necessary and sufficient condition to have Positive Assortative Matching (PAM) for any distribution of types is*

$$f_{xy} + f_{hy}h_x^* > 0 \quad \forall (x, y, V) \quad (7)$$

where $h_x^* = \frac{\partial h^*}{\partial x}$ and $h^* = h^*(x, y, V)$ is the equilibrium hours function.²⁴

Proof: see Appendix B.1. ■

Condition (7) has an intuitive interpretation. The term f_{xy} is standard in the literature and captures complementarities between types x and y in production. If f_{xy} is positive and large, high types x produce relatively more with high types y , pushing towards a PAM allocation. The second term, $f_{hy}h_x^*$, is new and captures the effect of the hours decision on the sorting patterns. When $f_{hy} > 0$, and $h_x^* > 0$ or $h_x^* < 0$ (but not too negative), the equilibrium allocation is pushed towards PAM. The condition says that if high type jobs produce relatively more with longer hours ($f_{yh} > 0$), and high-skilled workers work more than their low-skilled counterparts ($h_x^* > 0$), PAM is more likely to arise in equilibrium (for a given f_{xy}). In this case, hours are a way for high-skilled workers to secure top jobs. If, on the other hand, low- x workers work sufficiently more than high- x workers (h_x^* is sufficiently negative), they can compensate for their lower ability and be matched with the high types y , as long as their hours input is particularly valuable in top jobs ($f_{yh} > 0$). The opposite intuition is at work when $f_{yh} < 0$. Importantly, if $f_{yh} = 0$, hours are equally valuable across jobs, so working more or less hours has no impact on sorting.

The key takeaway from Proposition 1 is that the endogenous labor supply decision can create forces towards or against PAM, beyond standard type-complementarities embedded in f_{xy} ; these forces crucially depend on whether longer hours are more productive in high type jobs (which determines f_{yh}). Condition (7) is expressed in terms of an endogenous object, h_x^* . The advantage of this condition is that it's simple and intuitive, but leaves the question of how to express positive sorting in this model purely in terms of primitives. For this purpose, we derive the following proposition:

Proposition 2. *A necessary and sufficient condition on primitives to have Positive Assortative*

on conditions derived from the Hessian of the problem, as in Eeckhout and Kircher (2018). I show this in Appendix B.1.

²⁴Throughout the text, we denote partial derivatives with lowercase letters, e.g. $f_x = \frac{\partial f}{\partial x}$, and similarly for cross-partial derivatives.

Matching (PAM) for any distribution of types is

$$f_{xy} [-(u_{cc}f_hf_h + 2u_{ch}f_h + u_cf_{hh} + u_{hh})] + f_{hy} [(u_{cc}f_xf_h + u_cf_{xh} + u_{hc}f_x)] > 0 \quad \forall (x, y, h, c) \quad (8)$$

The opposite inequality provides a condition for Negative Assortative Matching (NAM). If utility is separable ($u_{ch} = 0$), then (8) simplifies to

$$f_{xy} [-(u_{cc}f_hf_h + u_cf_{hh} + u_{hh})] + f_{hy} [(u_{cc}f_xf_h + u_cf_{xh})] > 0 \quad \forall (x, y, h, c) \quad (9)$$

Proof: see Appendix B.1. ■

Inspecting closely this condition reveals further insights into the model. The first term on the left hand side of (8) is composed of the type-complementarity term f_{yx} , and the sum of terms $-(u_{cc}f_hf_h + 2u_{ch}f_h + u_cf_{hh} + u_{hh})$ which from optimality we know is positive.²⁵ Therefore, the sign of the second term is crucial to determine the allocation: If the second term $f_{hy}(u_{cc}f_xf_h + u_cf_{xh} + u_{hc}f_x)$ is negative and large enough, the allocation will be NAM (even if $f_{xy} > 0$). For example, if $f_{hy} > 0$, the key term becomes $(u_{cc}f_xf_h + u_cf_{xh} + u_{hc}f_x)$, which captures the key trade-off between income and substitution effects that drive the hours choice and with it, the equilibrium sorting. The first term ($u_{cc}f_xf_h$) is a by-product of the skill and hours premium f_xf_h combined with income effects coming from preferences, u_{cc} , and is therefore negative. The larger the curvature in utility (the more negative u_{cc}), the more negative this term and the less likely that high-skilled individuals will work longer hours. The term f_{xh} captures the incentive to work longer hours for high-skilled workers and is akin to a type-dependent substitution effect. The last term, $u_{hc}f_x$, captures the complementarity between hours worked and consumption. It follows that the sign of $(u_{cc}f_xf_h + u_cf_{xh} + u_{hc}f_x)$ is determined by the relative strength of income effects embedded in preferences on one hand, and hours complementarities (captured by $f_{xh} > 0$) on the other.

Hours Choice As the previous paragraph made clear, the key endogenous outcome that shapes equilibrium sorting is the hours choice by workers' skill type. The following proposition characterizes this choice explicitly.

²⁵This is because $-(u_{cc}f_hf_h + 2u_{ch}f_h + u_cf_{hh} + u_{hh})$ corresponds to the second order optimality condition of the hours problem.

Proposition 3. *In equilibrium, high-skilled workers choose higher hours ($h_x^* > 0$) if*

$$-f_h f_x \left(-\frac{u_{cc}}{u_c} + \frac{u_{ch}}{u_h} \right) + f_{xh} > 0 \quad \forall (x, y, h, c) \quad (10)$$

Proof: see Appendix B.1. ■

Condition (10) highlights once more that at the heart of the model lies a key trade-off between income effects in preferences, acting through $u_{cc} < 0$, and technology, acting through f_{xh} , which push h_x in opposite directions. As we have seen in Proposition 2, the same forces directly affect equilibrium sorting - and can reinforce or push against PAM. These opposing forces are similar to those arising in a static labor supply model.²⁶ The following corollary shows when the Proposition 3 holds in a wide class of commonly used preference specifications.

Corollary 1. *Assume that utility is separable in consumption and hours ($u_{hc} = 0$), then condition (10) simplifies to*

$$(u_{cc} f_h f_x + u_c f_{hx}) > 0 \quad \forall (x, y, h, c) \quad (11)$$

or, equivalently

$$(f(x, y, h) - V) \left(-\frac{u_{cc}}{u_c} \right) < \frac{f_{xh}(f(x, y, h) - V)}{f_x f_h} \quad \forall (x, y, h, c) \quad (12)$$

It follows that if $f - V$ is log-supermodular in (x, h) , then condition (12) is satisfied if the relative risk aversion coefficient $-c \frac{u_{cc}}{u_c} < 1$.

Proof: see Appendix B.1. ■

From the previous corollary, it is clear that $-c \frac{u_{cc}}{u_c} < 1$ is a sufficient condition for hours worked to be increasing in workers' attributes, x . A direct implication is that when $-c \frac{u_{cc}}{u_c} > 1$, sufficiently strong complementarities in production between skills and hours are required for the condition to be satisfied. This is of particular interest when considering preferences of the CRRA form, where $-c \frac{u_{cc}}{u_c}$ is simply given by a parameter σ . It follows that in this widely used preference class, assuming $\sigma > 1$ ($\sigma < 1$) is enough to determine whether income effects are stronger (weaker) than substitution effects, and hence

²⁶The key distinguishing feature here is that substitution effects can be heterogeneous across skills, i.e. the marginal product of one more hour worked is higher across skills (this happens when $f_{xh} > 0$).

whether a higher wage pushes towards working less (more). In other words, hours-skill complementarities in production can reinforce incentives to work long hours for high type workers even with large income effects.²⁷

3.3 Equilibrium Properties and Comparative Statics

We now illustrate the model's properties assuming a specific functional form for production and utility functions. We will assume a nested CES for technology

$$f(x, y, h) = A \left(\beta(\alpha x^\rho + (1 - \alpha)y^\rho)^{\frac{\gamma}{\rho}} + (1 - \beta)h^\gamma \right)^{\frac{1}{\gamma}} \quad (13)$$

and a [MaCurdy \(1981\)](#) utility function

$$u(c, h) = \frac{c^{1-\sigma}}{1-\sigma} - \psi \frac{h^{1+\frac{1}{\theta}}}{1+\frac{1}{\theta}}. \quad (14)$$

The advantage of this production technology is that it allows for flexible complementarity patterns, and allows for changes in the skill premium f_x and changes in skill-job complementarities f_{xy} through changes in α and ρ , as well as changes in hours-complementarities f_{xh}, f_{yh} through changes in γ .²⁸

3.3.1 Technological Change

We want to characterize how different types of technological change allowed in the model affect equilibrium incomes $w(x)$ and hours $h(x)$.²⁹ Figure 4 plots these outcomes for several values of γ . When γ gets more negative, hours turn from decreasing to increasing in skill type x (left panel), despite the large income effects operating in the background. In fact, income $w(x)$ becomes steeper in skills: better types are not only matched to better jobs, but they also work more, therefore increasing their marginal valuation in the market. In sum, a decrease in γ leads to higher types working relatively more hours, and because of that, getting rewarded more for their skills.

²⁷When utility is such that consumption and labor are complementary in consumption so that $u_{ch} > 0$, condition (10) is more likely to hold; the reason is that complementarities between hours and consumption can drive up the incentives to work for high wage workers.

²⁸Changes in hours-complementarities will be interpreted as Hours-Biased Technical Change in the quantitative application.

²⁹For these comparative statics exercises, and consistently with the quantitative application, we will consider points in the parameter space where PAM is satisfied for any type distribution, so the matching function $\mu(x)$ is unchanged along these specifications.

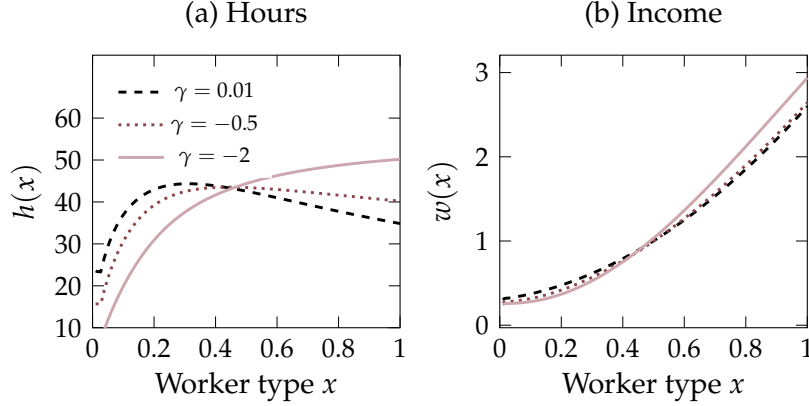


Figure 4: Equilibrium policy functions for alternative values of γ .

Notes: Income $w(x)$ is in thousands. Hours are weekly. The remaining parameter values are those corresponding to year 2010 in the estimated model (see Figure 8).

How do these effects compare with changes in the return to skill? It is useful to start from an increase in the weight of skill in production (α), but we show results for both ρ and α (in Figure 5 for α and Figure B.1 for ρ). In both cases, hours become decreasing in skills for large values of α and for more negative values of ρ , and this effect is driven by high-skilled working less.³⁰ The reason is precisely the income effects operating in the background.³¹ In sum, both changes in returns to skill (through α and ρ) and hours-complementarities (through γ) increase income disparities across workers, but they do so with opposite effects on their labor supply choices. In Appendix B.2, we provide a decomposition of these comparative statics further illustrating the role of sorting in driving these results.

3.3.2 The Role of Preferences

We conclude our comparative statics exercises section by showing how preferences can interact with technological changes to drive the equilibrium relationship between hours and skills in the cross-section, summarized here by the slope of (log) hours over (log) skills. Figure 6 plots this slope over a range of values of σ , for two values of the technology parameter γ . For either value of γ , the slope is decreasing, initially positive when $\sigma = 0$ and then turning negative for large enough values of σ . Interestingly, the value of σ required to make the slope negative is much higher when γ is lower. In other words, when σ is large, whether the slope is negative or positive depends on whether hours are

³⁰In fact, the effect for low-skilled is much more muted.

³¹The difference between ρ and α is that they change the equilibrium income function $w(x)$ in different ways: ρ mostly increases convexity, while α makes the income function steeper in x throughout the domain.

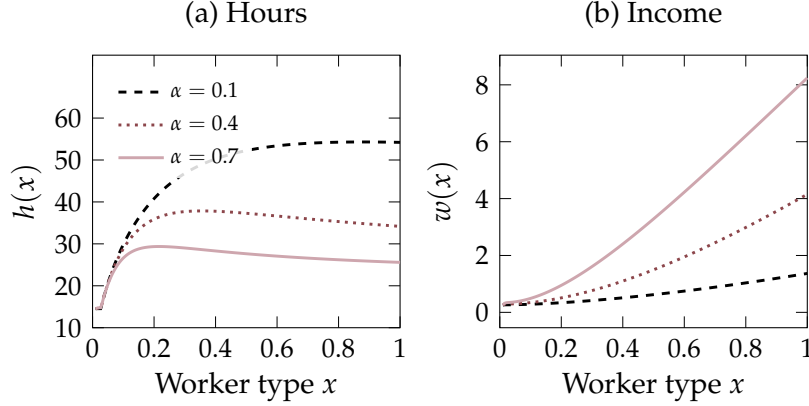


Figure 5: Equilibrium policy functions for alternative values of α .

Notes: Income $w(x)$ is in thousands. Hours are weekly. The remaining parameter values are those corresponding to year 2010 in the estimated model (see Figure 8).

sufficiently complementary to skills and jobs in production. When $\gamma = 0.01$, so that hours and the skill-job bundle are essentially Cobb-Douglas, a $\sigma > 1$ generates a negative slope; in this case, the model behaves similarly to a textbook model where $\sigma > 1$ implies income effects larger than substitution effects; when γ is such that hours and skills-jobs are Hicks complements (in the Figure, $\gamma = -1$), the slope shifts upward for any given level of σ .³²

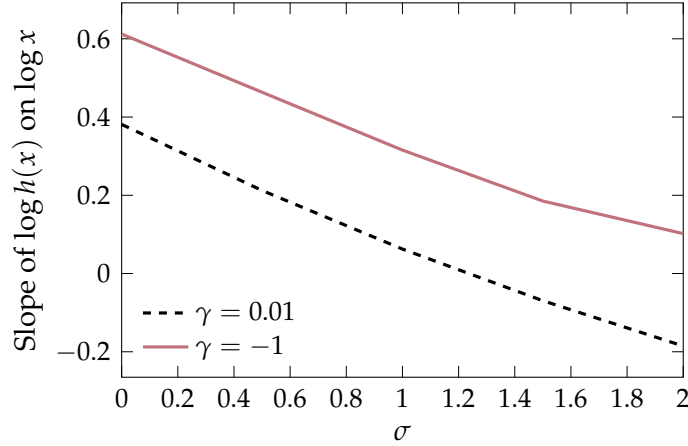


Figure 6: Slope of $\log h(x)$ on $\log x$ as a function of σ .

Notes: The figure plots, as a function of the consumption curvature parameter σ , the slope coefficient from a population-weighted OLS regression of $\log h(x)$ on $\log x$ across worker types, with one line per value of γ . The remaining parameter values are those corresponding to year 2010 in the estimated model (see Figure 8).

³²This result is reminiscent of Berger, Herkenhoff, Jeong and Mongey (2025), which show that the equilibrium sorting in a simplified version of their model depends on the relative values of σ and the worker-job complementarity terms.

3.4 Special Cases and Relationship to the Literature

The purpose of this section is to discuss how the framework relates to existing models, deriving the latter as special cases whenever possible.

Neoclassical Macro Frameworks The first link is the one with neoclassical growth frameworks with endogenous labor supply. Recently, [Boppart and Krusell \(2020\)](#) propose a general class of preferences where income effects exceed substitution effects. In their framework, the central object to be characterized is the Marginal Rate of Substitution (MRS) between consumption and hours, $MRS = -\frac{u_h}{u_c} = w$.³³ In this paper, the MRS condition is pair-specific and takes the form

$$MRS(c(x), h(x)) = -\frac{u_h(x)}{u_c(x)} = f_h(x, \mu(x), h(x)) \quad (15)$$

Equation (15) already highlights that in this framework, hours are chosen to balance a preference term MRS and a technology term f_h . The comparison with [Boppart and Krusell \(2020\)](#) can be made directly from (15). In the homogeneous-job benchmark, the marginal product of hours is common across workers; By contrast, in this assignment environment the return to an extra hour is match specific. Sorting therefore amplifies the substitution margin higher up the job ladder and provides a countervailing force to income effects.

Efficiency Units Efficiency units of labor are often used to model differences in labor inputs in macro-labor models. The common feature of these models is that the job cares about the labor h (hours) input scaled by the efficiency (or quality) of the input x , i.e. the jobs care about xh . In the current setting, production would take the form $f(x, y, h) = \tilde{f}(xh, y) = \tilde{f}(z, y)$ where z is efficiency units of labor. It can be shown that in this setting, for PAM it is sufficient to have $\tilde{f}_{zy} > 0$. The intuition behind this is that in equilibrium, high-skilled workers always choose enough hours so that efficiency units are higher than low-skilled workers (even if the latter work more). It follows that if those units are more valued in better jobs ($\tilde{f}_{zy} > 0$), high-skilled workers will always be allocated to those jobs. In other words, in this special case the hours' role in driving sorting is very limited due to the exact technological substitutability of hours and skills.

In the context of technological change and inequality, a prominent framework featuring efficiency units is [Krusell *et al.* \(2000\)](#). In that framework, the labor input of skilled

³³In [Boppart and Krusell \(2020\)](#), the MRS takes the form $MRS = -\frac{u_h}{u_c} = c^{\frac{1}{1-\nu}} q(hc^{\frac{\nu}{1-\nu}})$ for some function $q(\cdot)$. The implication is that on a balanced growth path, if consumption c grows at a rate $g^{1-\nu}$ and hours at a rate $g^{-\nu}$, the argument of $q(\cdot)$ stays constant and the MRS grows one for one with the wage, implying income effect is stronger than the substitution effect.

workers s in production is given by $s = \psi H$, where H is the total hours input and ψ captures the efficiency of the worker (e.g. human capital), and similarly for low-skilled workers l . In Appendix B.3.1, I show that even extending this framework to include hours in production in a possibly non-linear way (so that $s = \psi g(h)$) and a labor supply decision still implies that who works more hours depends only on the preference parameter σ , unless $g(h)$ is exogenously imposed to be type-specific ($g_s(h) \neq g_l(h)$).³⁴

It is also worth discussing the way the current framework differs from [Krusell *et al.* \(2000\)](#) and other works that model technological change as an exogenous shift in the price of equipment capital. In a production function as in (13), such changes can be represented by shifts in A, α, β . The additional generality of this framework is that it also allows for technical changes represented by changes in ρ, γ , which govern the returns to hours relative to skill across jobs; these changes then determine which jobs have the steepest returns to hours and, because of sorting, which workers are assigned to such jobs in equilibrium.

Hours and Skills as a Bundle This framework nests the assignment structure of a few models in the literature where skills and hours are ‘bundled’ together; in these models, only the bundle $g(x, h)$ enters the production function, so that production becomes $f = \tilde{f}(g(x, h), y)$. This class of models has been used frequently in the optimal taxation literature (see [Scheuer and Werning \(2017\)](#) or [Ales and Sleet \(2016\)](#)) but also recently in the literature on marriage and labor market sorting ([Calvo *et al.*, 2024](#)). With separable utility, it is easy to show³⁵ that Condition (9) reduces to $\tilde{f}_{gy} [-g_x u_{hh} + u_c \tilde{f}_g (g_h g_{xh} - g_x g_{hh})] > 0$; hence, if $\tilde{f}_{gy} > 0$, so that better jobs value effective labor more, a sufficient condition for positive assortative matching is $g_{hh} < \frac{g_h}{g_x} g_{xh}$.³⁶

It is also instructive to specialize the condition for hours to be increasing in skills, Proposition (3), to this case. Under separable utility, hours are increasing in skills if and only if $\frac{g_{xh}}{g_x g_h} > \tilde{f}_g \left(-\frac{u_{cc}}{u_c} \right) - \frac{\tilde{f}_{gg}}{\tilde{f}_g}$.³⁷ Hence, in these models, an upward-sloping hours schedule requires sufficiently strong complementarity between skills and hours inside the bundle.³⁸ In sum, because jobs value only the scalar $g(x, h)$, the bundle restriction therefore mechanically ties job-skill and job-hours complementarities. The technology used in the

³⁴In Appendix B.3.1, I provide a formal argument and characterization of how this framework relates to [Krusell *et al.* \(2000\)](#) and formally show the statements in this section.

³⁵See Appendix B.3.2.

³⁶A similar result arises in a model with sorting and taxation but with risk-neutral workers, as proven by [Vereshchagina \(2021\)](#).

³⁷If, in addition, $\tilde{f}$ is concave in the bundle, $\tilde{f}_{gg} \leq 0$, then $g_{xh} > 0$ is necessary for hours to increase with skills.

³⁸More generally, the bundle restriction implies that both sorting and hours depend on how $g(x, h)$ combines skills and hours, while ruling out any direct job-specific trade-off between them, since $\frac{f_h}{f_x} = \frac{g_h}{g_x}$ is independent of y .

quantitative section, instead, can separately govern worker-job sorting and the return to long hours in a match, allowing the model to separate SBTC, which encompasses worker-job complementarity, from HBTC, which changes hours-match complementarity.³⁹

Multiplicatively Separable Hours Frameworks such as [Erosa, Fuster, Kambourov and Rogerson \(2022\)](#) feature heterogeneity in the effects of working hours by occupation and comparative advantage of workers across occupations, extending [Roy \(1951\)](#): earnings e in occupation j take the form $e = a_j \cdot g_j(h)$, where a_j is the occupation-specific productivity of a worker and g_j the occupation-specific function that governs the elasticity of earnings to hours in that occupation. In the present framework, this corresponds to a multiplicatively separable technology of the form $f(x, y, h) = \bar{f}(x, y)z(y, h)$, where $\bar{f}(x, y)$ captures comparative advantage across worker-job pairs and $z(y, h)$ the occupation-specific returns to hours. Under this production structure and separable CRRA preferences,⁴⁰

$$\frac{f_{xh}(f - V)}{f_x f_h} = \frac{f - V}{f} \leq 1,$$

so that by Proposition (3) and Corollary (1), $h_x > 0$ is impossible for any $\sigma \geq 1$.⁴¹ This result affects the properties of the allocation. In the multiplicatively separable case, PAM is obtained when

$$\bar{f}_{xy}z(y, h) + \bar{f}_x z_y(y, h) + (\bar{f}_y z_h(y, h) + \bar{f} z_{hy}(y, h))h_x > 0.$$

Therefore, whenever better jobs also have steeper returns to hours, so that $f_{hy} = \bar{f}_y z_h + \bar{f} z_{hy} > 0$, the hours component pushes away from PAM for the empirically relevant case $\sigma > 1$, since in that case $h_x < 0$. A PAM allocation then requires a sufficiently strong direct sorting force, captured by $\bar{f}_{xy}z + \bar{f}_x z_y$, relative to the negative hours component.⁴² In that sense, the multiplicative separability can generate PAM only through the direct sorting force, effectively shutting down the mechanism at the core of the general model, namely that sufficiently strong hours complementarities can overturn the income effect and make high-skill workers work more.

³⁹This technology also allows changes over time in hours premia holding sorting forces fixed, and changes in sorting forces holding the hours technology fixed, which will be an important property in the counterfactuals.

⁴⁰See Appendix B.3.3 for details.

⁴¹More precisely, $\sigma > 1$ implies $h_x < 0$, while $\sigma = 1$ implies $h_x \leq 0$, with equality only if $V = 0$.

⁴²This mirrors the mechanism analyzed by [Vereshchagina \(2021\)](#) in the context of progressive taxation, where sufficiently strong direct complementarity preserves PAM under progressive taxation. More precisely, Proposition 4 in her paper requires $z(y, h)z_{hh}(y, h) \leq z_h(y, h)^2$ and $f_{xy} \geq (f_x/f_h)f_{hy} \geq 0$. When $f_y > 0$, the latter is equivalent to requiring that the Hicks elasticity of complementarity between x and y , $ff_{xy}/(f_x f_y)$, weakly exceeds that between h and y , $ff_{hy}/(f_h f_y)$.

4 Estimation

The goal of this section is to estimate the production function, as well as the skill and job types distributions, and assess the quantitative relevance of the mechanism in the paper. The model is estimated year-by-year for the years 1979-2018, a period of rapid technological advancements in the workplace and the period where the hours-wage correlation reversed sign and showed the largest increases. Moreover, this has been a period where the decline in average hours worked for the U.S. has slowed down and actually slightly reversed (see Figure 1), thus calling for an additional force beyond preferences to explain long-run behavior of hours worked. The purpose of this section is therefore precisely to identify one such force and examine its implications.

4.1 Functional Forms

Production Function I assume the same production function used in the comparative statics section, given by (13).⁴³ This gives us five technological parameters to be estimated year-by-year: $A_t, \alpha_t, \beta_t, \rho_t, \gamma_t$. Section 4.2 provides an identification argument for these parameters.

Preferences I make use of the same utility function (MaCurdy, 1981) as in the comparative statics section, given by (14). To calibrate this utility function, the values for σ and θ are taken from the literature, and ψ is normalized to 1. In particular, σ is set to 1.4, which is in the mid-range of values from the literature that aims at matching the overall decline in hours worked, both across time (Boppart and Krusell, 2020) and across the development path (Bick *et al.*, 2022b). The curvature in hours disutility cost θ is set to 0.5, a value well within the range of commonly used values to calibrate intensive-margin elasticities (see Keane (2011) for a survey).⁴⁴ Perhaps the most important feature of this calibration is that the curvature in consumption, σ , being above 1 implies that the substitution effect is being dominated by the income effect.⁴⁵

⁴³Note that this is a slightly more general version of the multiplicatively separable production function, of the form $f(x, y, h) = A(\alpha x^\rho + (1 - \alpha)y^\rho)^{\frac{\beta}{\rho}} h^{1-\beta}$. As noted for example in Chade and Lindenlaub (2022), the latter is perhaps one of the most commonly used production functions for empirical applications, and it is a special case of the production function I employ (in particular, it can be obtained by letting $\gamma \rightarrow 0$).

⁴⁴Notice that the calibration I adopt for the utility function is very close to available estimates of the same functional form from cross-sectional studies; in particular, Heathcote *et al.* (2014) employ a similar specification as in (14) and find values of σ and θ of 1.71 and 0.46, respectively.

⁴⁵This specification has the additional feature that leisure and consumption are separable, i.e. $u_{ch} = 0$. This formulation is convenient in this paper since - as the ultimate objective is to assess the role of technological complementarities in explaining patterns of hours inequality - it effectively shuts down complementarities between hours and consumption (or income) coming from the utility function, and hence isolates

Distributions Both the skills and the jobs distributions, $Q(x)$ and $G(y)$ are assumed to be unobserved, and hence they need to be estimated.⁴⁶ In particular, skills x and jobs y are assumed to follow log-normal distributions, with time-dependent parameters, given by $\log\mathcal{N}(a_t^x, b_t^x)$ and $\log\mathcal{N}(a_t^y, b_t^y)$, respectively. A similar approach has been used extensively in the literature, see e.g. [Lise *et al.* \(2016\)](#). A key advantage of this approach is that it allows for a year-by-year estimation of the model parameters without the need for data on (observable) distributions, which are rarely available at yearly frequency. The flexibility of this strategy therefore allows for an analysis of the underlying trends behind the parameter estimates, rather than only changes between two points in time.⁴⁷

4.2 Moments and Identification

Data CPS-ORG is used as the main sample, since it is available for the whole period of interest and it contains information on hours worked, earnings, and hourly wages. When computing the estimation targets, I will restrict the focus to the population of employed males.⁴⁸ I focus on the male population only because the analysis abstracts from the participation margin.

Moments I estimate the model by Simulated Method of Moments ([Pakes and Pollard \(1989\)](#), [McFadden \(1989\)](#)). In practice, I pick a set of moments m to identify the set of model parameters θ . For every year, the estimation procedure uses a combination of a global and a local search algorithm to seek for the parameter vector θ_t that minimizes the weighted square distance between model moments $m_t(\theta_t)$ and data moments $\bar{m}_t$:

$$\min_{\theta_t} (\bar{m}_t - m_t(\theta_t))' \Omega_t (\bar{m}_t - m_t(\theta_t)),$$

where Ω_t is the weighting matrix.⁴⁹ [Table 1](#) contains a summary of the targeted moments. For the details on the estimation algorithm, see [Appendix C.1](#).

the role of technology in explaining the data.

⁴⁶I interpret y in my model as jobs rather than firms. This is motivated by the application of the model, and specifically the technological changes at the center of the mechanism. This is a common interpretation in sorting models trying to understand rising inequality, see e.g. [Lindenlaub \(2017\)](#).

⁴⁷Analyzing these trends (rather than just points in time) is useful because SBTC and HBTC need not have accelerated (or decelerated) during the same years, and in general, because these trends contain valuable information on the identification of the production parameters, as will become clear in the identification argument presented later on.

⁴⁸For a full description of the data used in the estimation and all the sample restrictions used, see [Appendix A.1](#). For details on the estimation algorithm, see [Appendix C.1](#).

⁴⁹Following the literature, I use a diagonal weighting matrix given by the inverse of the (bootstrapped) variances of the empirical moments, i.e. $\Omega = \text{diag}(\widehat{\text{Var}}(\bar{m}))^{-1}$.

Table 1: Targeted Moments

Notation	Description
$E(w)$	Avg. Hourly Wages
$w90/w50$	90/50 Perc. Wage Ratio
$w90/w10$	90/10 Perc. Wage Ratio
$E[h \mid d^w = 9]$	Avg. hours worked in the 9th wage decile
$E[h \mid d^w = 5]$	Avg. hours worked in the 5th wage decile
$E[h \mid d^w = 2]$	Avg. hours worked in the 2nd wage decile
$std(w)$	Std of (log) wages

Identification I now discuss how each moment is related to the parameter to be estimated, with the understanding that the parameters are jointly estimated. The TFP level A informs the average income level in the model. The parameter β informs average hours worked in the 5th wage decile, as it determines the importance of hours in production (compared to job/skill) through $1 - \beta$. The mechanism is that β directly shifts up (or down) the marginal product of an hour worked, f_h . The parameters α , ρ and γ crucially determine the income and hours inequality in the model; however, key to identification is that they do so differently, as the comparative statics in the previous section have shown; in fact, while increases in α (and similarly, decreases in ρ) tend to reduce average hours for high wage workers, the opposite is true for decreases in γ . The assumed preference specification, and in particular the fact that income effects are stronger than substitution effects in preferences ($\sigma > 1$) is crucial to this result.⁵⁰ Moreover, α and ρ inform inequality at different points in the income distribution. The former increases overall inequality, while the latter governs the convexity in the wage function. Finally, the means of the worker and job type distributions a_t^x and a_t^y are normalized at 0 for all t . The worker type dispersion parameter b_t^x is normalized to 1, and we estimate the job type dispersion parameter b_t^y ; this argument is motivated by the fact that in this setting, only the ratio of these two parameters is identified.⁵¹ The parameter b_t^y is estimated yearly together with the production function parameters, and directly informs dispersion in wages, captured by the standard deviation of log wages.

⁵⁰We target mean hours in wage deciles 2, 5, and 9 to capture the lower, central, and upper parts of the wage distribution while avoiding the two outermost deciles, which are especially sensitive to hours measurement error and alternative trimming choices; see [Boppart et al. \(2024\)](#). The remaining deciles, including the two outermost groups, are not targeted and are examined as model-validation moments in [Figure C.1](#).

⁵¹In the positive assortative assignment equilibrium, the matching function is defined by $\mu(x) = G^{-1}(Q(x))$. Under lognormality, $Q_x(x) = \Phi\left(\frac{\log x}{b_t^x}\right)$, $G_y^{-1}(u) = \exp(b_t^y \Phi^{-1}(u))$, where $\Phi(\cdot)$ is the CDF of a normal distribution; therefore $\mu_t(x) = \exp\left(b_t^y \Phi^{-1}\left(\Phi\left(\frac{\log x}{b_t^x}\right)\right)\right) = \exp\left(\frac{b_t^y}{b_t^x} \log x\right) = x^{b_t^y/b_t^x}$. Hence the matching function depends on (b_t^x, b_t^y) only through the ratio b_t^y/b_t^x .

With this identification strategy, there are in total six parameters to be estimated: five from the production function ($A, \alpha, \beta, \gamma, \rho$) and one from the distributions of worker and job types (b^y). Table 2 contains a summary of the parameters to be estimated.

Table 2: Estimated parameters (time-varying)

Parameter	Meaning
A_t	Total Factor Productivity (TFP)
β_t	Weight of skills-job in production
α_t	Weight of skill in production (relative to job type)
γ_t	Substitution between hours and skill/job
ρ_t	Substitution between skill and job types
b_t^y	Variance of distribution of job types rel. to skill types

Measurement Error In order to align the model and data moments' construction, and in particular alleviate concerns about division bias (Borjas, 1980), we augment the model with measurement error, and assume that hours are reported with noise. In particular, we assume that - denoting the true value of hours in the model in year t , H_t - we observe two noisy hours measures with multiplicative (log-normal) reporting errors,

$$\begin{aligned}\log H_t^{\text{obs}} &= \log H_t + \varepsilon_{h,t}, & \varepsilon_{h,t} &\sim \mathcal{N}\left(-\frac{1}{2}\sigma_h^2, \sigma_h^2\right), \\ \log H_t^{\text{obs},w} &= \log H_t + \varepsilon_{w,t}, & \varepsilon_{w,t} &\sim \mathcal{N}\left(-\frac{1}{2}\sigma_h^2, \sigma_h^2\right),\end{aligned}$$

with $\text{corr}(\varepsilon_{h,t}, \varepsilon_{w,t}) = \rho_{hw}$.⁵² Hours-based moments are computed in the model from H_t^{obs} . Observed hourly wages are instead constructed by division $W_t^{\text{obs}} = Y_t / H_t^{\text{obs},w}$ where $H_t^{\text{obs},w}$ denotes the hours measure used in the wage denominator and Y_t is earnings. The advantage of this simple procedure is that the model-implied OLS relationship between hours and wages becomes a close counterpart of its empirical analogue, because it mimics the use of two distinct hours measures to mitigate the division bias problem (Bick *et al.*, 2018; Heathcote *et al.*, 2010), while leaving the equilibrium solution unchanged.⁵³ The value of σ_h^2 is set to 0.03.⁵⁴ We also set $\rho_{hw} = 0$, implying independent errors in reporting for the two hours measures.⁵⁵

⁵²The mean shifts $-\sigma_h^2/2$ ensure $E[H_t^{\text{obs}} \mid H_t] = H_t$, so measurement error affects dispersion but not average hours.

⁵³For consistency, we adopt an analogous procedure in cleaning the data used for estimation, and use two different hours measures available in CPS, as explained in the data paragraph.

⁵⁴ σ_h^2 induces an equivalent variance of measurement error in (log) wages, since $\log W_t^{\text{obs}} = \log Y_t - \log H_t - \varepsilon_{w,t}$, and hence $\text{var}(\log W_t^{\text{obs}} - \log W_t) = \sigma_h^2$.

⁵⁵A value of $\rho_{hw} = 0$ eliminates mechanical division bias from the model-implied cross-sectional rela-

4.3 Results

Model Fit The model and data moments targeted in the estimation are plotted in Figure 7. The model, despite being parsimonious, fits the data well. The model matches well the hours moments despite the hours choice being deterministic and the only source of randomness being measurement error. The ability of the model to match wage inequality measures is also reassuring in light of the counterfactual exercises of the next section.

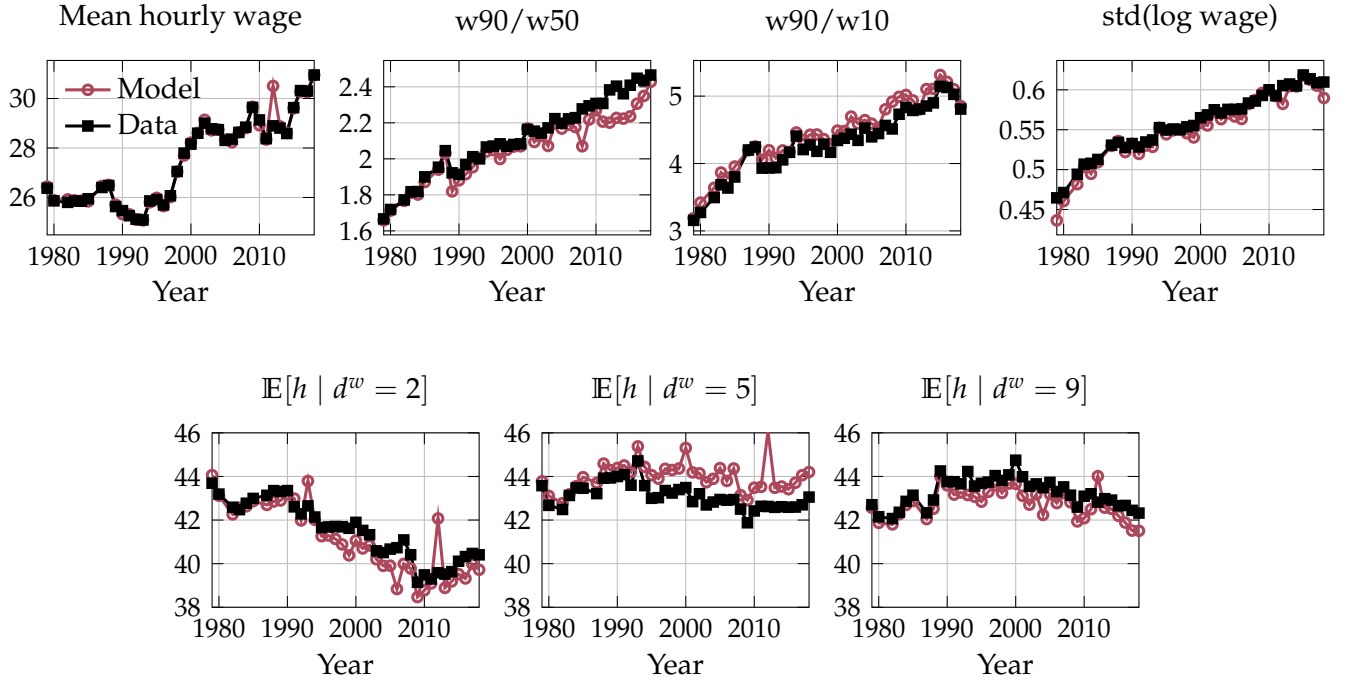


Figure 7: Model vs. data: targeted moments.

Notes: Black squared line plots the evolution of data moments, while red circled line plots model moments.

Estimates The estimated parameters are presented in Figure 8. The estimates reveal several interesting patterns, some new to this model, and some common given previous work focused on understanding the determinants of inequality in U.S. First, there has been an increase in the marginal product of skill in production, captured by α : from 0.17 (s.e. 0.01) to 0.23 (s.e. 0.002). Furthermore, complementarity between skills x and jobs y rose, i.e. ρ has decreased from -0.83 (s.e. 0.26) to -7.74 (s.e. 0.17), with most of the decrease concentrated in the first decades.⁵⁶ These patterns are at least qualitatively similar

relationship between hours and wages $\hat{\beta}$, while leaving active classical attenuation from measurement error in the denominator.

⁵⁶The targeted moments are imputed as raw data in the estimation algorithm, and therefore the parameter estimates inherit the unavoidable randomness in the year-to-year movements of data targets.

to what the literature on wage inequality in U.S. has found.⁵⁷ Furthermore, the estimated model reveals that the complementarity between skills/jobs and hours has also increased significantly: γ decreased, from a value of 0.04 (s.e. 0.02) to -0.51 (s.e. 0.04). These estimates imply that the economy has moved from a world that is close to Cobb–Douglas in the skill-job bundle and hours to one in which hours are much less substitutable with that bundle. As a result, long hours are rewarded disproportionately at the top of the skill and occupation distribution: extra hours matter more when supplied by the workers in the highest-value matches.⁵⁸ Finally, the estimates reveal an increase in the dispersion of job types relative to worker types which, given our normalization, is represented by an increase in b^y .⁵⁹ This means that the matching function $\mu(x)$ has become steeper, so that the same differences in worker types translate into larger differences in assigned job quality.⁶⁰

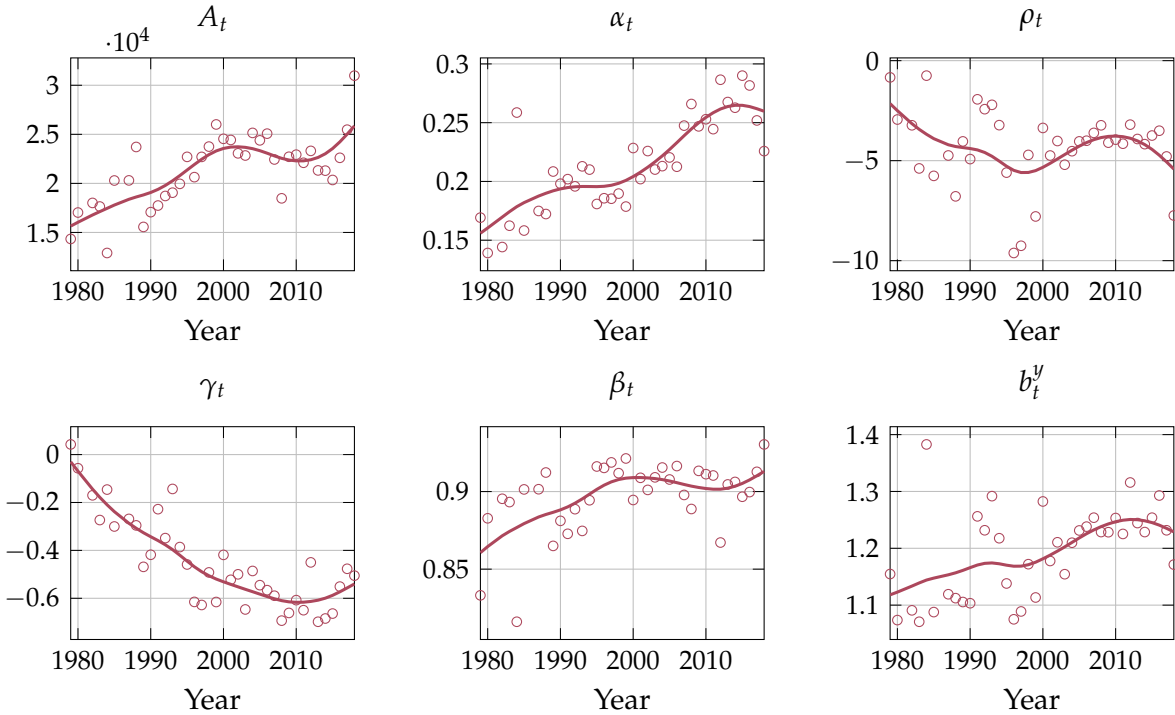


Figure 8: Estimated parameters by year.

Notes: Red circles plot the raw series and the red line plots the HP-filtered series, obtained with $\lambda = 100$.

⁵⁷For example, estimating a version of her model with one-dimensional skills, [Lindenlaub \(2017\)](#) finds that returns to skill have increased (see Table 18 in her paper).

⁵⁸In turn, the value of the parameter β , the weight of hours vs skills/jobs in production only showed a relatively mild increase over the time period considered.

⁵⁹Under the interpretation that y are occupations or bundles of tasks, this can be read as an increase in occupational heterogeneity, or a widening in the distribution of jobs available in the economy and larger differences across jobs in their productivity.

⁶⁰The estimation diagnostics and details on standard error computations are reported in [Appendix C.1](#).

Validation Figure 9 plots a few untargeted moments in the model and in the data as a simple and transparent validation exercise. The model captures well both the level and the trend in the cross-sectional elasticity between hours and wages (left panel). Moreover, it captures well both the increase and then decrease in the fraction of workers working more than 50 hours per week (middle panel). Finally, the right panel shows that the model captures a significant fraction of the dispersion in hours worked. On average, the model predicts between 55% and 60% of the overall dispersion (depending on the year considered). This is expected, since the model does not feature preference heterogeneity and all variation comes from either different work incentives or measurement error. To further examine the model's untargeted moments, a series of additional validation exercises that complement Figure 9 is presented in Appendix C.2: Figure C.1 shows the model's ability to closely track average hours worked by wage decile, despite only three of those deciles being targeted in estimation; Figures C.2 and C.3 compare, respectively, the incidence of long hours across wage deciles and the weekly-earnings distribution. Overall, these additional checks provide reassurance about the model's quantitative ability to match the moments of interest in the hours and income distribution, which is a key check in light of the counterfactual exercises in the next section.

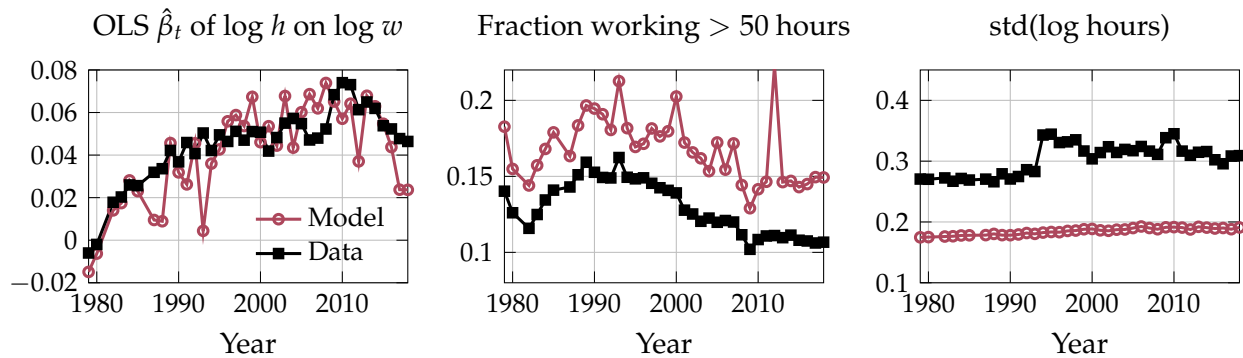


Figure 9: Model vs. data: untargeted moments.

Notes: Black squared line plots the evolution of data moments, while red circled line plots model moments.

5 Application: Income Inequality and the Role of Hours

The estimated model lends itself to a natural counterfactual exercise, which aims at looking at the evolution of income inequality in U.S. by explicitly taking into account the role of hours and sorting, and how technological change shapes this role.

5.1 Counterfactuals

Rise in Income Inequality In the following counterfactuals, we fix one set of estimated parameters to their initial level, and analyze the evolution of inequality in this counterfactual economy by feeding the model with all other parameters as estimated over time, effectively shutting down the channels of interest to gauge their importance. Skill-Biased Technological Change (SBTC) is referred to as changes in (α, b_y, ρ) , capturing increases in the return to skill x , broadly defined.⁶¹ An advantage of using a broad definition of SBTC is that we can immediately compare it with the effects of HBTC, which are by definition defined as changes in γ . The effects of shutting down technological changes on the overall rise in income inequality - defined as the variance of (log) income - are depicted in Figure 10.

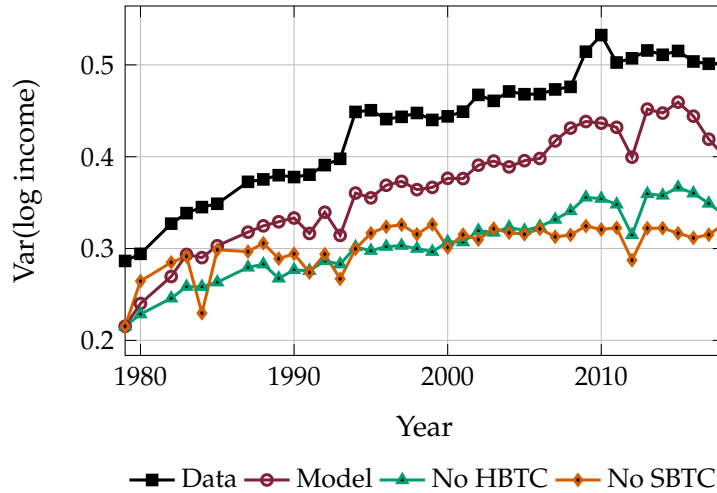


Figure 10: Variance of log income over time.

Notes: The figure displays the evolution of income inequality in the data ("Data"), in the estimated model ("Model"), and the counterfactual model evolution keeping γ at its initial estimated level ("No HBTC"), and keeping (α, b_y, ρ) at their initial estimated levels ("No SBTC").

The estimated model underpredicts the level of income inequality due to the fact that it only accounts for a fraction of the dispersion in hours, but it captures the evolution very well. Shutting down either SBTC or HBTC decreases the rise in income inequality.

⁶¹Increases in α raise the marginal product of skill x ; similarly, a lower ρ captures increases in complementarities between workers and jobs, while increases in b^y capture a stretching of the job-productivity distribution, which magnifies the gap in matched job quality between high- and low-skilled workers. We thus define SBTC in a broad sense to encompass both shifts in the production technology and shifts in the relative importance of high- y jobs. Both notions have been used in the literature to capture changes in the return to skill over time. Changes in production technologies are closer to frameworks like [Krusell *et al.* \(2000\)](#), while changes in b^y are akin to occupation-demand notions of SBTC as in [Autor and Acemoglu \(2011\)](#) or [Costinot and Vogel \(2010\)](#).

ity, thus showing the importance of both types of technological changes for its evolution, with HBTC explaining about 34% of the model-implied rise in inequality. This exercise demonstrates the quantitative importance of HBTC in explaining the rise in income inequality, but leaves the question of what causes this increase, which is what the next set of counterfactuals show.⁶²

Income Inequality: Decomposition Taking as our measure of inequality the variance of (log) income, we have the decomposition:

$$\text{var}(\log(\text{income})) = \text{var}(\log(\text{wage})) + \text{var}(\log(\text{hours})) + 2 \text{cov}(\log(\text{wages}), \log(\text{hours})) \quad (16)$$

A key advantage of this model is that rising income inequality can be studied in all its three components, since in this setting all three components are endogenous outcomes of the model. Exploiting the decomposition in (16), the next Figure shows the channels through which SBTC and HBTC affect total inequality.

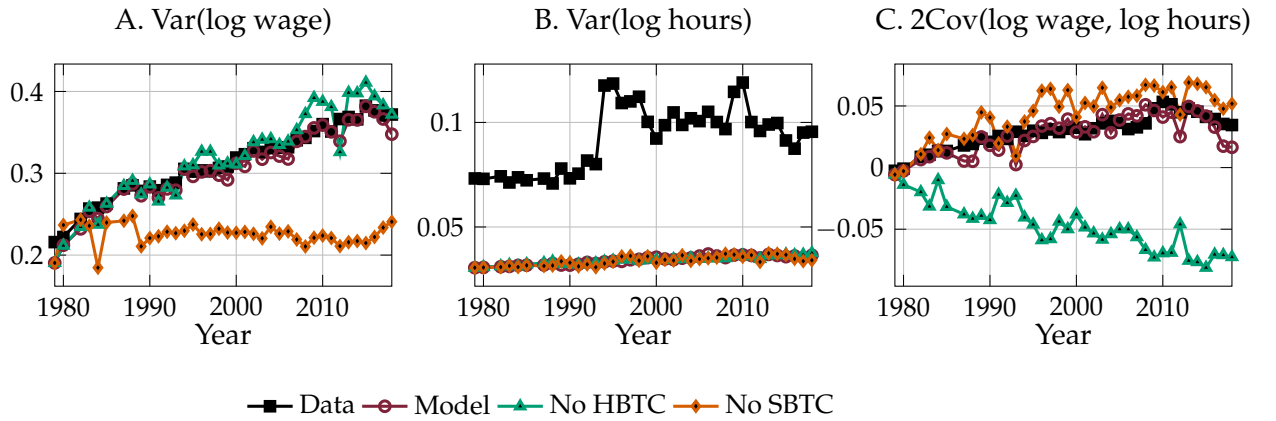


Figure 11: Rise in Income inequality: decomposition.

Notes: The figure displays the three components of income inequality in (16) (Panels A, B and C, respectively). For each panel, the figure plots the actual evolution of each component in the data (“Data”), the evolution in the estimated model (“Model”), and the evolution in the counterfactual model keeping γ at its initial estimated level (“No HBTC”), and keeping (α, b_y, ρ) at their initial estimated levels (“No SBTC”).

The results are plotted in Figure 11. Several results emerge. Focusing on Panel A., we notice that SBTC is the main driver of the increase in the variance of wages. Quantitatively, the effect of HBTC on the variance of wages is small since increases in the returns to

⁶²SBTC explains roughly 40% of the model-implied rise in inequality. The share of the rise explained by HBTC is computed as the fraction of the model-implied increase in inequality between 1979 and 2018 that disappears when γ is kept at its 1979 level. Formally, $\text{Share}_{\text{HBTC}} = \frac{(I_{2018}^{\text{model}} - I_{1979}^{\text{model}}) - (I_{2018}^{\text{no HBTC}} - I_{1979}^{\text{no HBTC}})}{I_{2018}^{\text{model}} - I_{1979}^{\text{model}}}$, where I is defined as $\text{Var}(\log(\text{income}))$ and analogously for SBTC.

working long hours are accompanied by the rising hours worked of this group, thus offsetting their effect on wages. Panel B. shows that neither SBTC nor HBTC had a material effect on hours dispersion. The reason this happens is that most of the dispersion is left unexplained by technological forces, and therefore these changes have minimal effects on the overall dispersion relative to the data. Finally, Panel C. (in line with the predictions of the theory) shows that the two estimated parameter changes have had opposite effects on the counterfactual relationship between hours and wages: without SBTC, the increase in the covariance between hours and wages would have been even higher than observed. Conversely, without HBTC, increases in the return to work long hours would be absent and hence the covariance would have decreased significantly due to the combination of SBTC and large income effects in preferences. Preferences, therefore, play a key role not only in explaining aggregate movements in hours but also in identifying the different effects that technological changes considered have on inequality.

The implication is that considering changes in the marginal product of skills x on the variance of wages hides important negative effects on inequality coming through the covariance term; likewise, changes in the marginal product of hours for certain types of workers is a new channel through which inequality increases, and becomes apparent only when the cross-sectional relationship between hours and wages is properly accounted for. This is a new result that has profound implications for our understanding of the deep causes of inequality, and complements the literature on Skill Biased Technical Change in that it shows that - in addition to the increased importance of skills in production - how these skills perform with more hours of work has first-order implications for inequality trends, both qualitatively and quantitatively.

5.2 Mechanism

We conclude the section with a summary of the mechanism, how it relates to existing work, and provide some suggestive evidence.

The Role of Sorting: Theory The model mechanism sheds light on the link between the nature of hours in production, summarized by the complementarity parameter γ , and the convexity of earnings in hours. This link operates through sorting. To see this, define $C(x, y) = (\alpha x^\rho + (1 - \alpha)y^\rho)^{1/\rho}$ and assume for simplicity that the assignment is positive assortative and equilibrium hours $h(x)$ are increasing in skill, so that the inverse $x = \eta(h) \equiv h^{-1}(x)$ is well defined. By optimality of the hours choice, the marginal return to hours evaluated along the equilibrium allocation is $m(h) \equiv f_h(\eta(h), \mu(\eta(h)), h) =$

$f_h(C(h), h)$. Differentiating this expression with respect to hours, we get

$$m'(h) = f_{hh}(C(h), h) + f_{hC}(C(h), h)C'(h). \quad (17)$$

Equation (17) characterizes the global returns to hours in this economy. The first term is the local curvature in the returns to hours and is negative whenever $f_{hh} < 0$; the second term can be positive or negative depending on the shape of the production function f_{hC} and the equilibrium assignment $C'(h)$. The term f_{hC} is positive whenever $\gamma < 1$; the equilibrium term $C'(h)$ means that high-skilled workers work more, and are assigned better jobs, so hours are higher in the skill-job bundle and hence $C'(h) > 0$.⁶³ Therefore, the combination of complementarities in hours together with sorting drives the term $f_{hC}(C(h), h)C'(h)$ and hence constitutes a force that raises $m'(h)$. In this light, sorting constitutes a micro-foundation for why earnings might have become more convex in hours worked (Goldin, 2014): because working more allows them to get better jobs, workers' incentives for long hours increase. This is especially true for top workers, because the return to their skills (and their hours) is magnified in top jobs. Moreover, this model also relates to the recent literature assessing the underlying heterogeneity in returns to hours across hours and occupations. In particular, sorting makes it possible for weak within-occupation returns to long hours and steep global returns to coexist, and thus identifies a new mechanism for which the observed earnings-hours profile need not coincide with the return to hours relevant for labor supply (Bick, Blandin and Rogerson, 2022a).

The Role of Sorting: Quantitative The best way to gauge the quantitative relevance of the mechanism is to ask how changes in γ interact in equilibrium with changes in gains to matching to drive up income inequality. To do so, we compute two counterfactual effects. The first is the effect of HBTC, computed on its own by starting from the 1979 economy and changing only γ to its estimated value in year t , while keeping all other parameters fixed at their 1979 values. Denoting the resulting level of income inequality (variance of log income) by I_t^{HBTC} , the effect of HBTC alone is

$$I_t^{HBTC} - I_0, \quad (18)$$

where I_0 is the model-implied value of inequality in the estimated 1979 economy. The interaction between changes in gains to matching and HBTC is gauged by computing a second counterfactual, given by the effect of HBTC in an economy in which the matching function and the complementarity between types undergo the estimated changes. Chang-

⁶³See Appendix B.4 for formal derivations.

ing b_y and ρ to their estimated values in year t , we obtain an economy with outcome $I_t^{\Delta\mu}$; additionally changing γ , we obtain an economy with outcome $I_t^{HBTC+\Delta\mu}$. The marginal effect of HBTC conditional on changes in gains to matching is therefore⁶⁴

$$I_t^{HBTC+\Delta\mu} - I_t^{\Delta\mu} \quad (19)$$

Comparing (18) and (19) reveals the nature of the interaction between changes in the matching function and HBTC across time.⁶⁵ Figure 12 plots these two objects. A positive interaction between HBTC and gains to matching emerges from the mid-1990's onwards, since the effect of the former in driving up inequality is much larger when the latter is also present. The intuition comes directly from the model's mechanism: because HBTC raises the returns to hours for high x , high y matches, when relative productivity in those matches increases, so does the effect of working one more hour relative to workers (and jobs) at the bottom of the distribution.⁶⁶

Evidence This section provides evidence that supports the mechanism. In particular, and following the analysis in Denning, Jacob, Lefgren and vom Lehn (2022) analysis, I decompose the returns of hours within jobs from the price of expected hours across occupations, which captures the across-occupations hours-wage elasticity. First, I run an individual-level regression of log hourly wages on log hours with detailed occupation fixed effects, which identifies the within-occupation elasticity. Second, I collapse the data to occupation-year means of log wages and log hours and estimate the corresponding across-occupation elasticity. The results are plotted in Figure A.4. The resulting series closely mirrors Denning *et al.* (2022) qualitative pattern obtained with U.S. Census data: the within-occupation elasticity remains close to zero throughout the sample, while the across-occupation elasticity is large and rises sharply over time. In other words, the rise in the return to hours is primarily an across-occupation phenomenon. This evidence provides support to the mechanism, in that it suggests that the allocation that long hours can grant to workers is an increasingly important component of the true returns to hours.⁶⁷

⁶⁴For notational convenience and brevity, we use $\Delta\mu$ to capture a change in the gains from matching, but it should be noted that only changes in b_y effectively change the shape of the matching function $\mu(x)$ (but not the fact that it's strictly increasing); changes in ρ , affecting the complementarity between x and y , amplify the gains from matching for a given matching function.

⁶⁵In fact, it follows that if $I_t^{HBTC+\Delta\mu} - I_t^{\Delta\mu} > I_t^{HBTC} - I_0$, then $\Delta\mu$ amplifies the effect of HBTC on I . If instead $I_t^{HBTC+\Delta\mu} - I_t^{\Delta\mu} < I_t^{HBTC} - I_0$, then $\Delta\mu$ dampens it.

⁶⁶As Figure C.4 in Appendix C.3 shows, this is due to the fact that HBTC is now reflected in increased wage dispersion and an increased covariance between hours and wages when both changes are present.

⁶⁷Moreover, this new evidence is consistent with the literature that estimates returns to long hours across occupations and skills: Goldin (2014), Cortés and Pan (2019), and Jang and Yum (2022) emphasize that some occupations disproportionately reward long and continuous hours, which is consistent with the estimated

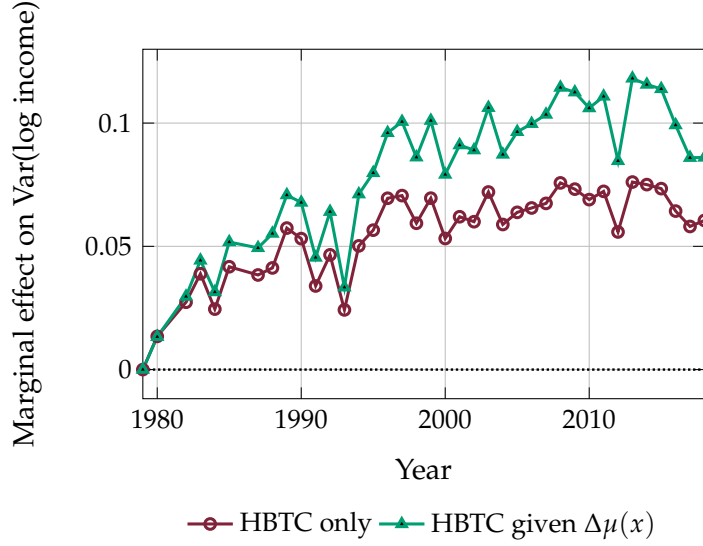


Figure 12: Interaction of HBTC with changes in $\mu(x)$.

Notes: The figure displays a) the increase in the variance of log income in the model economy when estimated changes in γ are fed into the 1979 estimated economy, leaving all other parameters unchanged, relative to the level of the variance of log income in 1979 ("HBTC only"); b) the increase in the variance of log income when both estimated changes in γ and in ρ, b^y are fed into the 1979 estimated economy relative to when only ρ, b^y are changing ("HBTC given $\Delta\mu$ ").

6 Discussion

I now discuss two settings in which the mechanism proposed in this paper might have additional implications. I conclude the section by discussing alternative explanations.

6.1 Implications

Hours Constraints and Misallocation The first setting is a stochastic extension of the model to study the misallocation effects of hours distortions. The extension replaces the deterministic assignment with an allocation of worker-job pairs and hours that not only depends on production complementarities and preferences, but also on EV-I type idiosyncratic shocks.⁶⁸ The extension is used to study a simple reduced-form distortion that raises the nonpecuniary cost of supplying hours, $\psi_\tau = (1 + \tau)\psi$, where $\tau \geq 0$ indexes the strength of the wedge (which can be interpreted as social norms, child-care related costs, commuting costs, or other nonpecuniary burdens). The full exposition of the model and

decline in γ .

⁶⁸The reference framework is Galichon, Kominers and Weber (2019), extended to have hours entering in production. For a recent application of a similar framework, see Bandiera, Kotia, Lindenlaub, Moser and Prat (2024).

the exercise can be found in Appendix D.1.

The exercise of interest is to compare the effect of changes in the wedge τ under two technology regimes, indexed by the strength of the hours-complementarity parameter γ . The main result is that the incidence of the wedge is highly unequal across workers. Income losses are largest for high-skilled workers, even though the direct reduction in long-hours caused by higher τ is more broadly distributed across the skill distribution. The wedge also reallocates top jobs away from top workers and toward lower-skill workers, because the effects of changing τ operate not only through the intensive margin of hours but also through the assignment of workers to jobs. These results seem particularly relevant when studying gender gaps and the misallocation of skilled women across jobs. If interpreted as unequal division of labor at home, the effects of a gender-specific wedge causing high-ability women to be allocated away from top jobs could be magnified by HBTC, resulting in higher losses in incomes and output - especially for workers whose long hours are most valuable in the market.⁶⁹ More generally, the extension suggests a potentially meaningful role for HBTC in amplifying exogenous differences in hours constraints, with heterogeneous effects across the skill distribution.

Labor Supply Elasticities The second setting relates to labor supply elasticities and optimal tax progressivity. Following the increase in income inequality in the U.S., the question is whether tax progressivity should respond to this increase, and if so, by how much (Heathcote, Storesletten and Violante, 2020b; Ferriere, Grübener and Sachs, 2025).⁷⁰ Therefore, it is interesting to look at how labor supply elasticities respond to HBTC in this setting, and how sorting affects these elasticities, both qualitatively and quantitatively.

To do so, in Appendix D.2, I show a comparison between compensated elasticities in this model and elasticities computed with fixed assignment (i.e., shutting down the effect that working more hours has on the allocation of workers to jobs), in the spirit of Scheuer and Werning (2017).⁷¹ The full elasticity then adds the assignment margin. Importantly, this gap itself depends on the technology governing the return to hours, which we have argued has changed over time due to HBTC.

Appendix D.2 shows that, in the top tail, the elasticity relevant once workers are allowed to move along the assignment schedule exceeds the fixed-assignment elasticity.

⁶⁹We fully pursue this extension in separate work, see Mantovani and Qiu (2025).

⁷⁰The literature has studied optimal taxation in the presence of superstar effects similar to those analyzed in this setting (Scheuer and Werning, 2017; Ales and Sleet, 2016), albeit with production functions different from the one used in the quantitative application of this paper.

⁷¹This distinction is useful because, in the present model, the same force emphasized in the mechanism section - namely that hours are more valuable in better matches - affects not only inequality but also the elasticity of earnings with respect to incentives.

Moreover, the average proportional adjustment is systematically larger in the estimated economy than in the no-HBTC counterfactual, particularly in recent decades. In other words, HBTC increases the extent to which assignment affects the relevant earnings elasticity. Under the estimated changes in the production function, this adjustment is quantitatively significant.

These findings suggest a possible implication for discussions of tax progressivity. In standard sufficient-statistics approaches, the key behavioral input is the earnings elasticity. In this setting, however, that object is not invariant to technology: it depends on how hours enter production and on the extent to which higher-hour workers are sorted into better jobs. Fully characterizing how progressivity should change over time is beyond the scope of the paper⁷², but these results suggest that the type of technological changes emphasized in this paper may make unadjusted within-job elasticities increasingly misleading in environments where hours and sorting are strongly intertwined.⁷³

6.2 Alternative Explanations

The objective of the paper is to propose and quantify a new mechanism to jointly account for cross-sectional and aggregate hours worked movements. The theory purposely abstracts from a number of additional features that have been shown to play an important role in labor supply. These features are discussed next.

Taxes and Transfers A more (or less) progressive tax-and-transfer system can affect hours through both disposable income and the net return to work. For example, under preferences for which the income and substitution effects of wage growth exactly cancel, SBTC can raise wages at the top without changing hours, while an expansion of transfers concentrated at the bottom can reduce the hours of low-wage workers. Conditional on the initial hours profile, this mechanism could in principle contribute to the reversal of the hours-wage relationship. However, this argument predicts constant hours at the top, while in the data, both average hours and the incidence of long workweeks rose signifi-

⁷²Establishing that tax progressivity should change in this setting would require solving the full normative problem, which is beyond the scope of the paper. The message here is that the technological changes emphasized in the paper alter the earnings-elasticity object entering sufficient-statistics tax calculations. While [Scheuer and Werning \(2017\)](#) show that superstar effects can be neutral conditional on the correct sufficient statistics, the stronger comparative-static cancellation result in that paper need not apply automatically here, because output depends separately on worker type, job type, and hours, and sorting and hours are jointly determined in equilibrium, as in [Vereshchagina \(2021\)](#).

⁷³[Vereshchagina \(2021\)](#) studies the effects of changes in tax progressivity on sorting when hours are relevant in production, finding that the reallocation of managers to jobs following tax reforms can have significant effects on output and inequality; these results further emphasize the importance of considering the sorting of workers to jobs when analyzing optimal tax rates.

cantly among high-wage workers.⁷⁴

Whether income tax progressivity has effectively increased in the US is an open question. Wu (2021) finds that income tax progressivity has declined in US. On the other hand, Heathcote *et al.* (2020b) show that once transfers are taken into account and progressivity is examined across the whole distribution, conventional measures of tax progressivity⁷⁵ have been stable from the 1980's onwards. Across countries and along the development path, Bick *et al.* (2022b) and Qiu and Russo (2026) estimate the degree of tax progressivity and show that GDP per capita is positively correlated with the degree of progressivity: richer countries exhibit a more progressive tax system. Overall, this evidence seems to suggest that, while changing progressivity certainly remains a driver for the decision to work longer hours, it leaves room for other mechanisms to explain the data.

Insurance A large literature distinguishes between permanent and transitory components of the wage process and studies their contribution to inequality. This distinction is particularly relevant once workers can save and smooth consumption: A rise in the variance of temporary wage shocks can increase the hours-wage correlation even when income effects are large, because workers can take advantage of temporary windows of opportunity by increasing their hours when wages are high.⁷⁶ Heathcote *et al.* (2014) show that insurable wage variation tends to make wages and hours move together, whereas uninsurable permanent variation can generate the opposite relationship when income effects dominate. Similarly, Boppart *et al.* (2024) show quantitatively that the degree of insurance can substantially affect the cross-sectional allocation of hours across workers.

Whether a similar mechanism can explain the observed trends is ultimately an empirical matter, as it depends jointly on whether wage inequality became increasingly transitory or insurable and on whether households' ability to insure wage variation changed over time.⁷⁷ Thus, while these considerations remain valid and plausible, introducing savings does not by itself generate the trend in the hours–wage relationship, and therefore the theory in this paper abstracts from this margin to theoretically isolate the role of the new mechanism proposed here.

⁷⁴Moreover, these preferences do not generate the long-run decline in hours through productivity growth alone, a key motivation for the preference specification used in this paper.

⁷⁵As is common in the literature, the measure of tax progressivity they focus on is the parameter that governs progressivity in the income tax function used in Heathcote *et al.* (2014).

⁷⁶More generally, the relevant distinction is how strongly wage variation passes through to consumption. For example, as emphasized by Boppart and Krusell (2020), access to assets can allow workers to smooth consumption and shift hours toward high-wage periods even when permanent productivity growth reduces hours in the long run.

⁷⁷Heathcote *et al.* (2014) indicate that the relative importance of insurable wage variation was broadly stable after the early 1980s. As noted in Aguiar and Bils (2015), several recent papers using administrative data highlight the importance of permanent income differences across individuals.

Human Capital Endogenous human-capital accumulation can also link hours and skills. For example, in [Michelacci and Pijoan-Mas \(2012\)](#), longer hours increase workers’ skills and improve their access to better jobs. To provide an alternative to HBTC, hours worked today must raise future skills and earnings, and this investment motive must strengthen over time. A common implication of life-cycle models where returns to hours are dynamic is that the hours response should be stronger among younger workers, who have a longer horizon over which to recoup the investment. In Appendix [D.3](#), we report age patterns in the rise in hours and the probability of working long hours within the top wage decile: Figures [D.3](#) and [D.4](#) show that average weekly hours rose across the age distribution, with larger increases among older workers, while the rise in the probability of working long hours was concentrated among workers aged 45 and above.⁷⁸ While these findings do not rule out a role for human-capital accumulation, they suggest that dynamic considerations are unlikely to be the main explanation for the rise in hours at the top.

7 Concluding Remarks

This paper has developed and quantified a new theory of sorting in which workers are assigned to jobs based on their skills and the hours they provide. The main mechanism emerging from the theory is that the allocation of workers to jobs through hours is a key component of the true earnings returns to hours. The quantitative analysis shows that this margin has become stronger over time due to pervasive technological advancements in the labor market. Empirical evidence based on [Denning *et al.* \(2022\)](#) supports the mechanism, and in particular the role of sorting as a key determinant of the returns to hours faced by workers, above and beyond the within-job returns. This echoes [Bick *et al.* \(2022a\)](#), who show that the market earnings-hours schedule cannot be inferred directly from the raw cross-sectional relationship between wages and hours; in this setting, the relevant return to hours also reflects how hours change the worker’s assignment to jobs.

In the final part of the paper, I consider a stochastic version of the model to show that wedges to working long or flexible hours can misallocate talent across jobs, in the spirit of [Hsieh, Hurst, Jones and Klenow \(2019\)](#). The broader message is that, once technology makes the returns to working time more heterogeneous across jobs, inequality and aggregate productivity increasingly depend not only on who possesses more skill, but also on who can supply the hours needed to be allocated to the jobs where those skills are most

⁷⁸[Alon *et al.* \(2019\)](#) provide complementary evidence, showing the absence of a systematic relationship across occupations between changes in returns to experience and changes in average hours.

valuable.

The quantitative analysis focuses on employed men and therefore abstracts from the extensive margin. This restriction may matter because nonemployment contributes directly to population-wide earnings inequality and changes in participation may alter the composition of workers observed in employment. These considerations are especially relevant at the bottom of the distribution: [Heathcote, Perri and Violante \(2020a\)](#) show that the decline in weeks worked among low-earning men is accounted for primarily by a decline in the fraction working, while [Birinci, Karabarbounis and See \(2026\)](#) emphasize the role of employment rates in the evolution of hours per person. By contrast, declining employment at the bottom does not mechanically account for the rise in hours and in the incidence of long workweeks among high-wage employed men, which is a key intensive-margin movement emphasized in this paper.⁷⁹

A natural extension would introduce nonemployment as an outside option and jointly determine participation, hours conditional on employment, and assignment across jobs. Because participation determines the distribution of worker types entering the matching market, such an extension could attribute part of the observed changes in wages and hours to selection rather than to technology.⁸⁰

While the focus of the paper is to introduce and quantify a new mechanism, all these considerations suggest that fully pursuing these extensions represents a fruitful avenue for future research.

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⁷⁹Of course, selection within the upper tail could still affect these moments, but would require a distinct change in the composition of high-wage workers.

⁸⁰This issue is likely to be most consequential at the bottom of the male distribution, where employment has changed the most, and in an extension to women, given the large secular changes in female participation.

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Online Appendix

Not for publication

A Data Appendix

A.1 Sample Construction

Overview The main dataset used in the analysis is the Current Population Survey (CPS). In particular, I use the Outgoing Rotation Group (ORG) due to the detailed information on earnings and hours worked.⁸¹ I adopt the following sample restrictions: I only consider males, aged 25-64. Multiple-job holders, self-employed, and workers with usual weekly hours below 10 are also excluded from the sample. The dataset is then trimmed of the bottom 1% of the usual hours distribution. All statistics are computed using the provided sample weights.

Hours Unless otherwise noted and stated explicitly, all hours statistics are computed using *hourslw*, which is hours worked last week across all jobs. Hours moments (means, dispersion, percentiles, and hours-by-wage-decile profiles) are computed using *hourslw* and the CPS-ORG weights. In addition, the hours-wage elasticity is computed year-by-year from a weighted regression of $\log(\text{hourslw})$ on $\log(rw)$ controlling for age and age squared.

Wages Real hourly wages are measured by the CEPR variable *rw* (real hourly wage in 2019 dollars) and are used as provided in the extracts. Wage dispersion is measured using the within-year standard deviation of residual log wages: for each year I regress $\log(rw)$ on a quartic in age and compute the weighted standard deviation of the residuals. Wage percentiles and wage-decile group assignments are computed using weighted inverse-CDF (step) quantiles based on the CPS-ORG weights.

A.2 Robustness and Additional Evidence

A.2.1 Hours-Wage Elasticity Trends

Figure A.1 reports six robustness checks for the hours-wage elasticity in Figure 3 (left panel). Panel (a) uses CPS-ASEC, while panels (b)–(f) vary the sample restrictions, hours definition, and controls in the CPS-ORG analysis.

Raw Hours and Wages By Wage Decile In figure A.2, I divide the sample of workers in CPS by wage decile (i.e., I compute the wage decile year by year) and plot the average

⁸¹The main advantage of using CPS-ORG is that questions on work and hours refer to current pay or usual hours, unlike CPS-ASEC, which uses as a reference period of the last week or last year.

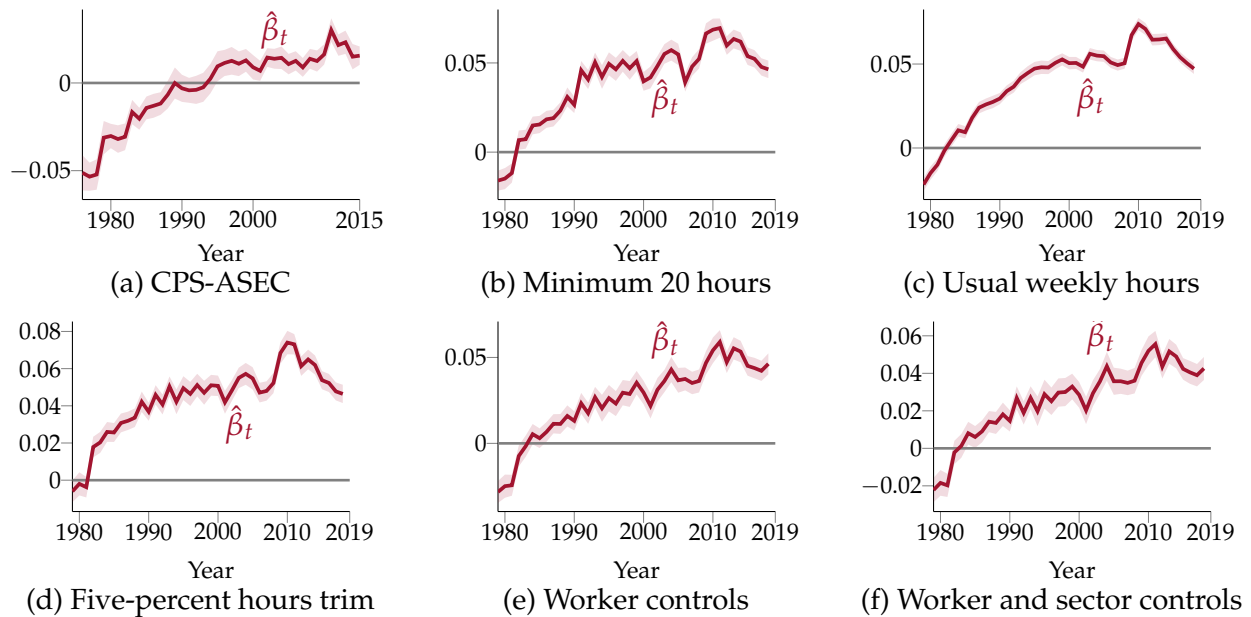


Figure A.1: Hours-wage elasticity: robustness checks.

Notes: Each panel reports the estimated coefficient from a weighted year-by-year regression of log hours on log hourly wages; shaded areas are 95% confidence intervals based on heteroskedasticity-robust standard errors. Panel (a) uses the CPS-ASEC sample and annual hours, following [Boppart et al. \(2024\)](#). It restricts the sample to men aged 25–54; excludes members of the armed forces, individuals outside the labor force, individuals with unemployment during the year, and observations in the bottom 1% of the within-year annual-hours distribution; and controls for age, age squared, education categories, and race. Panels (b)–(f) use the baseline CPS-ORG sample of men aged 25–64 and retain its age and age-squared controls. Relative to that baseline, panel (b) excludes workers with usual weekly hours below 20; panel (c) uses usual weekly hours on the main job (*uhouse*) instead of hours worked last week; panel (d) trims the bottom 5% rather than the bottom 1% of the hours distribution; panel (e) adds five education categories (less than high school, high school, some college, college, and advanced degree) and four race/ethnicity categories (White, Black, Hispanic, and Other); and panel (f) adds those worker controls plus fixed effects for 14 harmonized broad industries derived from EPI’s time-consistent *mind16* classification. All regressions use the provided CPS sample weights.

hours worked by each decile. To avoid cluttering, I only plot average hours worked for the top, the bottom, and the middle decile. The figure clearly shows that workers with the lowest wages (black line) decreased their hours worked; workers with highest wages (red line) increased their hours, albeit the increase is concentrated in the 1980’s and 1990’s.

ATUS Next, I turn to another dataset to inspect whether trends in hours worked and leisure by skill category are unique to CPS. I use the American Time Use Survey (ATUS), already used for studies regarding leisure trends, e.g. [Aguiar and Hurst \(2007\)](#)⁸².

Taking data from [Boppart and Ngai \(2021\)](#) (see their paper for details on data construction), I plot average hours worked on the market and leisure hours by skill category (I

⁸²Following [Aguiar and Hurst \(2007\)](#), the data for the pre-2003 comes from America’s Use of Time datasets and the 1975–1976 Time Use in Economics and Social Accounts. See their paper for details.

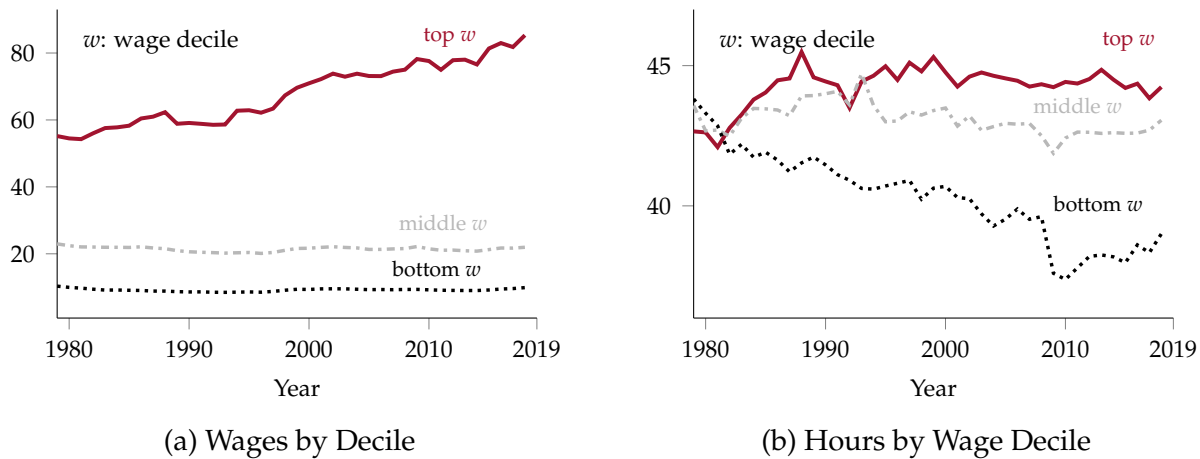


Figure A.2: Wages and Hours by Wage Decile.

Notes: Panel (a) plots mean hourly wages and panel (b) mean weekly hours for workers in the lowest-wage decile (bottom 10 percent), the fifth wage decile (between the 40th and 50th percentiles), and the highest-wage decile (top 10 percent). Wage deciles are recomputed separately in each year. Source: CPS-ORG.

approximate skill by years of education). I plot the evolution of market hours and leisure hours by educational category in Figure A.3 (left and right panel, respectively). From the beginning of the sample until roughly 1985, workers of each education category consume more leisure hours and less market hours, with changes parallel along educational categories. Starting from 1985, highly educated workers diminish their leisure hours and increase their market hours; low-educated workers continue to provide lower market hours in all subsequent years.

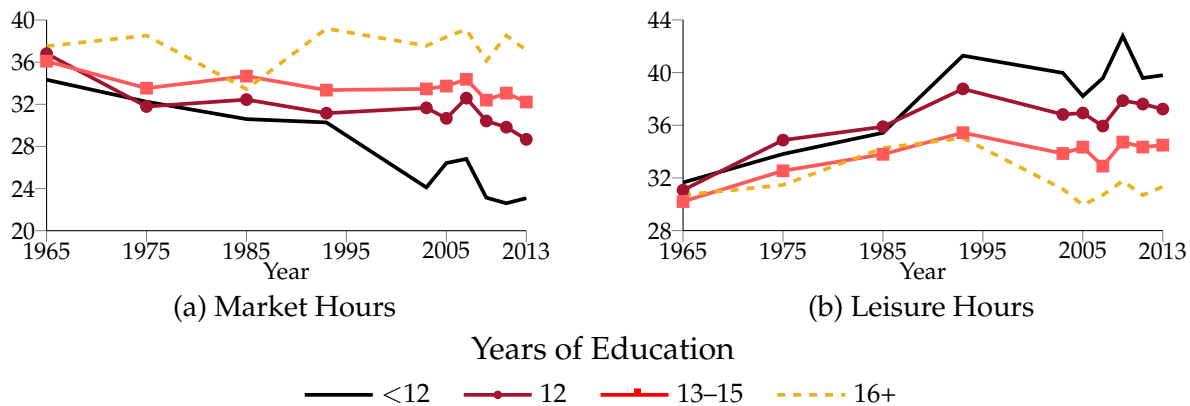


Figure A.3: Market and leisure hours by education group.

Notes: The left panel plots market hours and the right panel plots leisure hours by education group, ATUS dataset. The four groups correspond to less than 12 years of education, 12 years, 13-15 years, and 16 or more years. Source: Boppart and Ngai (2021) and Aguiar and Hurst (2007).

A.3 Empirical Evidence

The following Figure replicates, using the CPS-ORG sample, an analogous analysis as in [Denning *et al.* \(2022\)](#). The Figure shows clearly that the within-occupation elasticity of wages relative to hours worked, as in their paper, is negative or around zero throughout the recent decades; the across-occupation elasticity, on the other hand, rises sharply and continuously throughout the period, suggesting that returns to hours are primarily a between-occupation phenomenon.

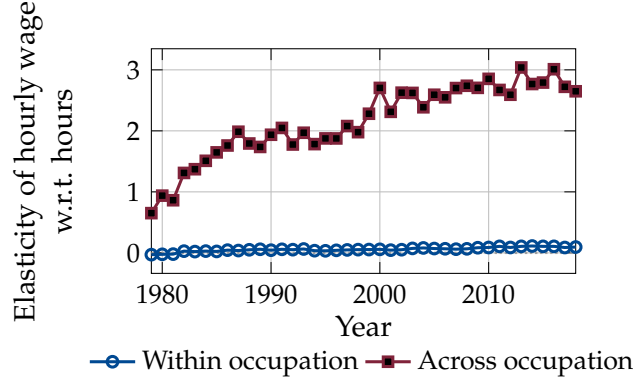


Figure A.4: Denning-style decomposition of the elasticity of hourly wages with respect to hours.

Notes: The figure plots the within-occupation component and the across-occupation component of the elasticity of log hourly wages with respect to log hours. The within component (blue circled line) is estimated from year-specific regressions of log hourly wages on log hours with 3-digit occupation fixed effects. The across component (brown squared line) is estimated from year-specific regressions of occupation-year mean log hourly wages on occupation-year mean log hours.

B Theory Appendix

B.1 Conditions for Assortative Matching

Proof of Proposition 1 As shown in [Chade *et al.* \(2017\)](#), under the regularity conditions in Theorem 3 in [Milgrom and Shannon \(1994\)](#), condition (6) is equivalent to the Spence-Mirrlees single crossing condition. In this setting, this means that the Marginal Rate of Substitution (MRS) $-\frac{\Phi_y}{\Phi_V}$ is increasing in x . We have:

$$\Phi_y = u_c f_y, \quad \Phi_V = -u_c$$

Therefore, $-\frac{\Phi_y}{\Phi_V}$ increasing in x implies that $f_y(x, y, h^*(x, y, V))$ is increasing in x . Then, differentiating $f_y(x, y, h^*(x, y, V))$ with respect to x , we directly obtain (7), which completes the proof. ■

Proof of Proposition 2 The first-order condition for the optimal hours choice in (5) is given by:

$$u_c(f(x, y, h) - V, h)f_h(x, y, h) + u_h(f(x, y, h) - V, h) = 0.$$

Define

$$\mathcal{F}(x, y, V, h) \equiv u_c(f(x, y, h) - V, h)f_h(x, y, h) + u_h(f(x, y, h) - V, h).$$

Then $h^*(x, y, V)$ satisfies the FOC

$$\mathcal{F}(x, y, V, h^*(x, y, V)) = 0. \quad (\text{B.1})$$

By the implicit function theorem,

$$h_x^* = -\frac{\mathcal{F}_x}{\mathcal{F}_h}. \quad (\text{B.2})$$

The numerator of (B.2) is

$$\mathcal{F}_x = u_{cc}f_xf_h + u_cf_{xh} + u_{hc}f_x. \quad (\text{B.3})$$

The denominator of (B.2) is

$$\mathcal{F}_h = u_{cc}f_h^2 + (u_{ch} + u_{hc})f_h + u_cf_{hh} + u_{hh}. \quad (\text{B.4})$$

Recalling that, from Proposition 1, PAM obtains if and only if

$$f_{xy} + f_{hy}h_x^* > 0 \quad (\text{B.5})$$

we can write h_x^* in (B.5) explicitly by substituting the numerator and denominator terms (B.3) and (B.4). This is equivalent to

$$f_{xy} - f_{hy} \frac{u_{cc}f_xf_h + u_cf_{xh} + u_{hc}f_x}{u_{cc}f_h^2 + (u_{ch} + u_{hc})f_h + u_cf_{hh} + u_{hh}} > 0.$$

Since the denominator is negative by the strict second-order condition⁸³, multiplying by it reverses the inequality. Therefore PAM obtains if and only if

$$f_{xy} \left[- \left(u_{cc}f_h^2 + 2u_{ch}f_h + u_cf_{hh} + u_{hh} \right) \right] + f_{hy} [u_{cc}f_xf_h + u_cf_{xh} + u_{hc}f_x] > 0. \quad (\text{B.6})$$

⁸³The second-order condition (SOC) for the hours choice is

$$u_{cc}f_h^2 + (u_{ch} + u_{hc})f_h + u_cf_{hh} + u_{hh} < 0.$$

for all (x, y, h, c) , which is the condition in the main text. The opposite inequality gives the condition for NAM. The simplified condition (9) is immediately obtained when $u_{ch} = 0$ is assumed. ■

Alternative derivation

We can obtain the same condition by following the joint-decision argument. Let

$$\Phi(x, y, h) = u(f(x, y, h) - V(y), h),$$

where $V(y)$ is the equilibrium payoff schedule for the job side. The first-order conditions with respect to h and y are

$$\Phi_h = u_c f_h + u_h = 0,$$

and

$$\Phi_y = u_c(f_y - V_y) = 0.$$

Since $u_c > 0$, the second condition implies $V_y = f_y$ along the equilibrium allocation. Differentiating the two first-order conditions with respect to x gives

$$\begin{pmatrix} \Phi_{hh} & \Phi_{hy} \\ \Phi_{yh} & \Phi_{yy} \end{pmatrix} \begin{pmatrix} h_x \\ \mu_x \end{pmatrix} + \begin{pmatrix} \Phi_{hx} \\ \Phi_{yx} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Using $V_y = f_y$ and the relevant derivatives (omitted for brevity)

$$\begin{pmatrix} u_{cc}f_h^2 + 2u_{ch}f_h + u_cf_{hh} + u_{hh} & u_cf_{hy} \\ u_cf_{yh} & u_c(f_{yy} - V_{yy}) \end{pmatrix} \begin{pmatrix} h_x \\ \mu_x \end{pmatrix} + \begin{pmatrix} u_{cc}f_x f_h + u_cf_{xh} + u_{hc}f_x \\ u_cf_{xy} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Solving for μ_x gives

$$\mu_x = \frac{u_cf_{hy}(u_{cc}f_x f_h + u_cf_{xh} + u_{hc}f_x) - u_cf_{xy}(u_{cc}f_h^2 + 2u_{ch}f_h + u_cf_{hh} + u_{hh})}{\begin{vmatrix} u_{cc}f_h^2 + 2u_{ch}f_h + u_cf_{hh} + u_{hh} & u_cf_{hy} \\ u_cf_{yh} & u_c(f_{yy} - V_{yy}) \end{vmatrix}}.$$

By optimality, the joint second-order condition implies that

$$\begin{vmatrix} u_{cc}f_h^2 + 2u_{ch}f_h + u_cf_{hh} + u_{hh} & u_cf_{hy} \\ u_cf_{yh} & u_c(f_{yy} - V_{yy}) \end{vmatrix} > 0.$$

which is to say the determinant of the Hessian of the joint hours and type decision needs

to be positive. Since $u_c > 0$, PAM, i.e. $\mu_x > 0$, is therefore equivalent to

$$f_{hy} (u_{cc}f_xf_h + u_cf_{xh} + u_{hc}f_x) - f_{xy} (u_{cc}f_h^2 + 2u_{ch}f_h + u_cf_{hh} + u_{hh}) > 0.$$

Thus this derivation delivers the same primitive condition as the main derivation, and it is more similar to the one used in [Eeckhout and Kircher \(2018\)](#). ■

Proof of Proposition 3 To prove the result, we use monotone comparative statics. Recall that the first order condition for hours choice is simply $u_cf_h + u_h = 0$. Then by monotone comparative statics, to prove that h^* weakly increases in the worker type x , we need that:

$$\frac{\partial(u_cf_h + u_h)}{\partial x} > 0 \quad (\text{B.7})$$

Hence we need that

$$u_{cc}f_xf_h + u_cf_{xh} + u_{ch}f_x > 0 \quad (\text{B.8})$$

Using the FOC $u_cf_h + u_h = 0$ into (B.8), and rearranging terms, we get that

$$-f_hf_x\left(-\frac{u_{cc}}{u_c} + \frac{u_{ch}}{u_h}\right) + f_{xh} > 0 \quad (\text{B.9})$$

which proves the result. ■

Proof of Corollary 1 Condition (11) directly follows from (10) after assuming $u_{ch} = 0$ and rearranging terms. To derive (12), start from (11) and multiply both sides by $f - V$ and rearrange terms. For the second part of the corollary, recall that log-supermodularity of a continuous, twice-differentiable function f in x, y implies:⁸⁴

$$f_{xy}f \geq f_xf_y$$

Therefore, log-supermodularity of the term $f - V$ in the corollary implies

$$f_{xh}(f - V) \geq f_hf_x \quad (\text{B.10})$$

or

⁸⁴Log-supermodularity is a stronger condition than supermodularity if $f_x, f_y \geq 0$.

$$\frac{f_{xh}(f - V)}{f_h f_x} \geq 1 \quad (\text{B.11})$$

Hence, if $f - V$ is log-supermodular, then the RHS in Corollary 1 is bigger than the LHS - and the condition is satisfied - if the LHS is lower than 1. Using the budget constraint of the household $c = w$ and the fact that $f - V = w$ implies that the LHS is simply $\left(-c \frac{u_{cc}}{u_c}\right)$ which is the coefficient of relative risk aversion. It follows that if the coefficient of risk aversion is lower than 1, the condition is satisfied - and hours worked are increasing in the worker attributes - as long as $f - V$ is log-supermodular, i.e. $\left(-c \frac{u_{cc}}{u_c}\right) < 1$ is a sufficient condition for $h_x^* > 0$. ■

B.2 Additional Figures: Allocation and Comparative Statics

Equilibrium Effects of lower ρ Figure B.1 plots the equilibrium effects of lower ρ , complementing the analysis in the main text.

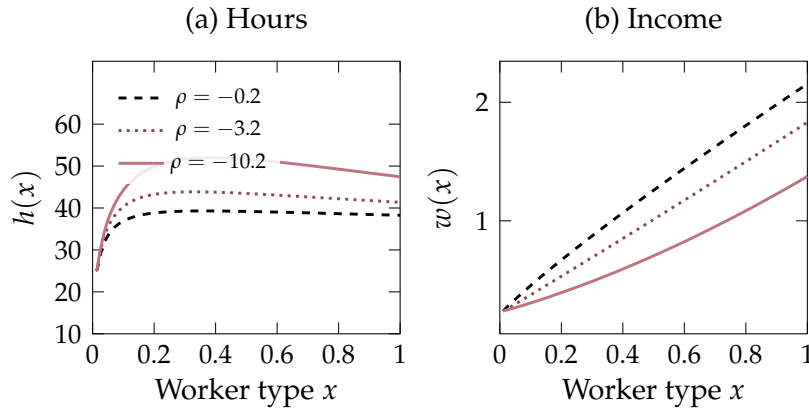
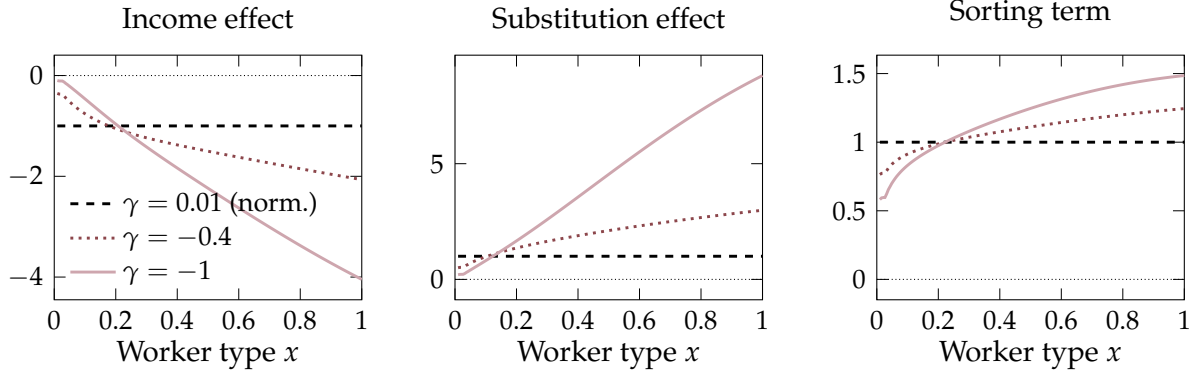


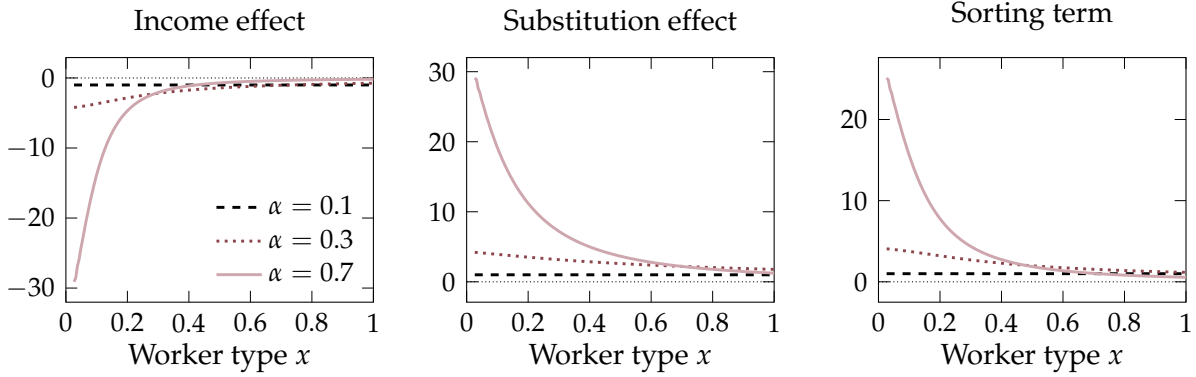
Figure B.1: Equilibrium policy functions for alternative values of ρ .

Notes: Income $w(x)$ is in thousands. The remaining parameter values are those corresponding to year 2010 in the estimated model (see Figure 8).

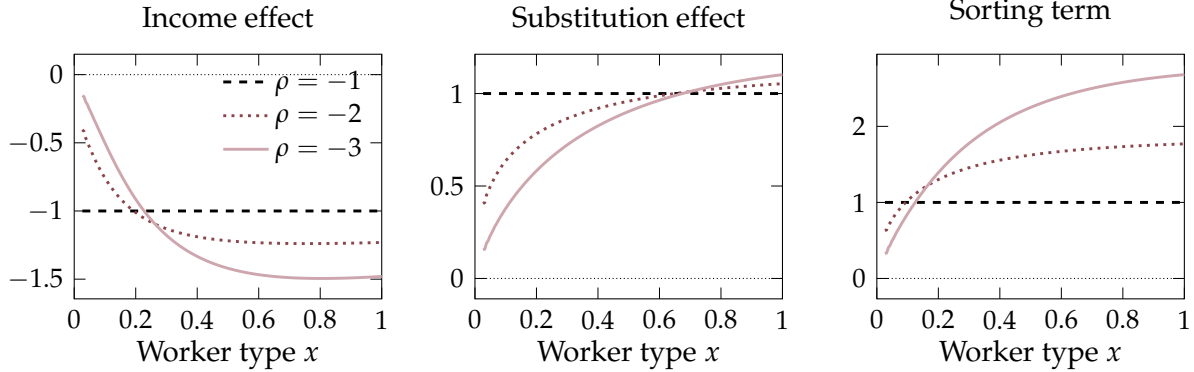
Decomposition of the PAM allocation We unpack the main comparative statics responses and illustrate the mechanism behind them. To do so, note that the terms in Condition 9: an income effect term $f_{yh}u_{cc}f_xf_h$, a substitution effect term $f_{yh}u_cf_{xh}$, and a sorting term $-f_{xy}(u_{cc}f_h^2 + u_cf_{hh} + u_{hh})$. We plot each of these terms (normalized by a baseline value of γ) separately to understand how changes in technology drive the equilibrium responses shown in Figure 4 and Figure 5. As Figure B.2a makes clear, a lower γ makes the income effect term more negative at the top of the skill distribution (left panel), while it increases the substitution effect term for high-skilled (especially relative to low-skilled). The third term shows that the sorting term is also pushed up by a lower γ , even though its effect is



(a) Alternative values of γ .



(b) Alternative values of α .



(c) Alternative values of ρ .

Figure B.2: Decomposition of (9) for alternative technology parameters.

Notes: Each row plots, along the equilibrium matching path $y = \mu(x)$, the three terms governing the sign of positive assortative matching in (9): the income effect $f_{yh}u_{cc}f_x f_h$ (left), the substitution effect $f_{yh}u_c f_{xh}$ (center), and the sorting term $-f_{xy}(u_{cc}f_h^2 + u_c f_{hh} + u_{hh})$ (right). In panel (a), the dashed, dotted, and solid lines represent $\gamma = 0.01$, -0.4 , and -1 ; in panel (b), they represent $\alpha = 0.1$, 0.3 , and 0.7 ; and in panel (c), they represent $\rho = -1$, -2 , and -3 . Each curve is normalized pointwise by the absolute value of the corresponding term at the dashed baseline, which is therefore flat at ± 1 ; the other curves show the force relative to that baseline. The remaining parameters are fixed at their 2010 estimated values (see Figure 8).

smaller in magnitude than the substitution effect.

The decomposition exercises for α and ρ are plotted in Figure B.2b and Figure B.2c, respectively. Similarly to the case of γ , a lower ρ makes the income effect more negative for high-skilled workers; the substitution effect, however, barely moves at the top and only changes a bit at the bottom. By contrast, as shown in the right panel, the sorting term increases, reinforcing the direct type-complementarity effect governed by ρ . Increasing α , on the other hand, operates mostly at the bottom of the distribution, where all terms are amplified. However, these effects jointly reduce the incentives to work at the top relative to the bottom, and therefore the hours function becomes more decreasing in skills as α increases, as shown in the main text.

B.3 Special Cases: Proofs and Derivations

B.3.1 Efficiency Units

Assignment in the simple efficiency units model First, we prove that in an efficiency-units version of this paper's framework, PAM simply depends on the sign of $\tilde{f}_{zy} > 0$. Suppose production takes the form

$$f(x, y, h) = \tilde{f}(z, y), \quad z \equiv xh.$$

Assume separable utility,

$$\bar{u}(c) - u(h), \quad c = \tilde{f}(z, y) - V,$$

with the usual properties that $u_h > 0$ and $u_{hh} \geq 0$. By Equation (7), PAM obtains if

$$f_{xy} + f_{hy}h_x^* > 0.$$

Under $f(x, y, h) = \tilde{f}(xh, y)$, we have that $f_x = h\tilde{f}_z$, $f_h = x\tilde{f}_z$, $f_{xy} = h\tilde{f}_{zy}$, $f_{hy} = x\tilde{f}_{zy}$. Hence the PAM condition becomes

$$\tilde{f}_{zy}(h + xh_x^*) > 0. \tag{B.12}$$

Now, defining equilibrium efficiency units

$$z^*(x, y) = xh^*(x, y).$$

Differentiating,

$$z_x^*(x, y) = h + xh_x^*.$$

Now, rewrite the worker's problem in terms of z :

$$\max_z \bar{u}(\tilde{f}(z, y) - V) - u(z/x).$$

The first-order condition is

$$\bar{u}_c(\tilde{f}(z, y) - V)\tilde{f}_z(z, y) = \frac{1}{x}u_h(z/x).$$

Define

$$H(z, x; y, V) \equiv \bar{u}_c(\tilde{f}(z, y) - V)\tilde{f}_z(z, y) - \frac{1}{x}u_h(z/x).$$

At the optimum, $H(z^*, x; y, V) = 0$. By the implicit function theorem,

$$z_x^* = -\frac{H_x}{H_z}.$$

Now

$$H_x = -\frac{\partial}{\partial x}\left(\frac{1}{x}u_h(z/x)\right) = \frac{u_h(z/x) + (z/x)u_{hh}(z/x)}{x^2} > 0.$$

Under the second-order condition, $H_z < 0$, implying $z_x^* > 0$. Combining the previous expressions,

$$h + xh_x^* = z_x^* > 0.$$

Hence a sufficient condition for PAM, from (B.12), is

$$\tilde{f}_{zy} > 0.$$

Next, we show that by itself, an extension to [Krusell *et al.* \(2000\)](#) (KORV) to have hours in production doesn't change the fact that preferences govern which skill type works more in equilibrium. Consider a framework as in KORV in which labor of type $i \in \{u, s\}$ (unskilled, skilled) enters production through efficiency units

$$L_i = x_i N_i g(h_i),$$

where x_i is worker efficiency, N_i is the mass of workers of type i , and $g(h)$ is common across types. Let the representative firm price one unit of effective labor of type i at p_i , so that earnings of type i are

$$e_i(h_i) = p_i x_i g(h_i) \equiv \omega_i g(h_i), \quad \omega_i \equiv p_i x_i.$$

This is still an efficiency-units model, since the firm only cares about the bundle $x_i g(h_i)$. Now suppose each type solves

$$\max_{h_i} u(c_i) - v(h_i) \quad \text{subject to} \quad c_i = \omega_i g(h_i),$$

and let $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$. The first-order condition is

$$u'(\omega_i g(h_i)) \omega_i g'(h_i) = v'(h_i).$$

or equivalently

$$\omega_i^{1-\sigma} g(h_i)^{-\sigma} g'(h_i) = v'(h_i).$$

Define

$$\Phi(h, \omega) \equiv \omega^{1-\sigma} g(h)^{-\sigma} g'(h) - v'(h).$$

At the optimum, $\Phi(h^*(\omega), \omega) = 0$. By the implicit function theorem,

$$\frac{dh^*}{d\omega} = -\frac{\Phi_\omega}{\Phi_h}.$$

Now,

$$\Phi_\omega = (1 - \sigma) \omega^{-\sigma} g(h)^{-\sigma} g'(h).$$

Hence the sign of Φ_ω is the sign of $1 - \sigma$. Under the second-order condition, $\Phi_h < 0$, so $\frac{dh^*}{d\omega}$ has the same sign as $1 - \sigma$. Thus, in the canonical decentralized KORV-with-hours benchmark with a common hours technology $g(h)$, who works more depends only on the preference parameter σ .⁸⁵

Embodied technical change inside the CES specification. We show how the current model can capture embodied technical change as in [Krusell *et al.* \(2000\)](#) through changes in A, α, β for given γ and ρ . The importance of this derivation is not only to clarify the meaning of these parameters in this context relative to papers where technological change is modeled as embodied in capital, but also to show the usefulness of allowing for flexible complementarity patterns, captured by γ and ρ . Consider the production function

$$f(x, y, h) = A \left[\beta (\alpha x^\rho + (1 - \alpha) y^\rho)^{\gamma/\rho} + (1 - \beta) h^\gamma \right]^{1/\gamma}.$$

⁸⁵Of course, this conclusion can change if the model is extended to have e.g. insurance motives, or if one imposes exogenous type-specific hour technologies $g_s(h) \neq g_u(h)$.

Suppose technical change acts on the job side through a multiplicative shifter q_t , so that production (13) becomes

$$f_t(x, y, h) = A \left[\beta (\alpha x^\rho + (1 - \alpha)(q_t y)^\rho)^{\gamma/\rho} + (1 - \beta)h^\gamma \right]^{1/\gamma}.$$

Define

$$\kappa_t = \alpha + (1 - \alpha)q_t^\rho, \quad \alpha_t = \frac{\alpha}{\kappa_t}, \quad 1 - \alpha_t = \frac{(1 - \alpha)q_t^\rho}{\kappa_t}.$$

Then

$$\alpha x^\rho + (1 - \alpha)(q_t y)^\rho = \kappa_t (\alpha_t x^\rho + (1 - \alpha_t)y^\rho),$$

so

$$(\alpha x^\rho + (1 - \alpha)(q_t y)^\rho)^{\gamma/\rho} = \kappa_t^{\gamma/\rho} (\alpha_t x^\rho + (1 - \alpha_t)y^\rho)^{\gamma/\rho}.$$

Hence

$$f_t(x, y, h) = A \left[\beta \kappa_t^{\gamma/\rho} (\alpha_t x^\rho + (1 - \alpha_t)y^\rho)^{\gamma/\rho} + (1 - \beta)h^\gamma \right]^{1/\gamma}.$$

Now define

$$D_t = \beta \kappa_t^{\gamma/\rho} + (1 - \beta), \quad A_t = A D_t^{1/\gamma}, \quad \beta_t = \frac{\beta \kappa_t^{\gamma/\rho}}{D_t}.$$

Then

$$1 - \beta_t = \frac{1 - \beta}{D_t},$$

and therefore

$$f_t(x, y, h) = A_t \left[\beta_t (\alpha_t x^\rho + (1 - \alpha_t)y^\rho)^{\gamma/\rho} + (1 - \beta_t)h^\gamma \right]^{1/\gamma}.$$

This shows that embodied technical change on the job side (q_t) can be mapped directly in this framework as time variation in A_t, α_t, β_t , holding ρ and γ fixed.

Why ρ and γ govern the return to hours relative to skill across jobs. Let

$$C(x, y) = (\alpha x^\rho + (1 - \alpha)y^\rho)^{1/\rho},$$

so that

$$f(x, y, h) = A \left[\beta C(x, y)^\gamma + (1 - \beta)h^\gamma \right]^{1/\gamma}.$$

Differentiating with respect to hours and skills,

$$f_h = A(1 - \beta)h^{\gamma-1} \left[\beta C(x, y)^\gamma + (1 - \beta)h^\gamma \right]^{1/\gamma-1}.$$

$$f_x = A\beta\alpha x^{\rho-1}C(x, y)^{\gamma-\rho} \left[\beta C(x, y)^\gamma + (1 - \beta)h^\gamma \right]^{1/\gamma-1}.$$

Hence

$$\log\left(\frac{f_h}{f_x}\right) = \log\left(\frac{1 - \beta}{\beta\alpha}\right) + (\gamma - 1)\log h + (1 - \rho)\log x + (\rho - \gamma)\log C(x, y).$$

Since

$$\log C(x, y) = \frac{1}{\rho} \log(\alpha x^\rho + (1 - \alpha)y^\rho),$$

we obtain

$$\frac{\partial \log C(x, y)}{\partial y} = \frac{(1 - \alpha)y^{\rho-1}}{\alpha x^\rho + (1 - \alpha)y^\rho}.$$

Therefore

$$\frac{\partial}{\partial y} \log\left(\frac{f_h}{f_x}\right) = (\rho - \gamma) \frac{(1 - \alpha)y^{\rho-1}}{\alpha x^\rho + (1 - \alpha)y^\rho}.$$

This shows that if $\rho \neq \gamma$, the return to hours relative to skill varies systematically across jobs.

B.3.2 Bundle models

Suppose utility is separable in consumption and hours, so that $u_{ch} = 0$, and let

$$f(x, y, h) = \tilde{f}(g(x, h), y).$$

Then, we have that $f_x = \tilde{f}_g g_x$, $f_h = \tilde{f}_g g_h$, $f_{xy} = \tilde{f}_{gy} g_x$, $f_{hy} = \tilde{f}_{gy} g_h$, $f_{xh} = \tilde{f}_{gg} g_x g_h + \tilde{f}_g g_{xh}$, $f_{hh} = \tilde{f}_{gg} g_h^2 + \tilde{f}_g g_{hh}$. Recall that, under separable utility,

$$h_x = \frac{u_{cc} f_x f_h + u_c f_{xh}}{\varphi},$$

where, because of optimality

$$\varphi = -u_{cc} f_h^2 - u_c f_{hh} - u_{hh} > 0.$$

Condition (9) can be written as

$$f_{xy}\varphi + f_{hy}(u_{cc} f_x f_h + u_c f_{xh}) > 0.$$

Substituting the expressions above gives

$$f_{xy}\varphi + f_{hy}(u_{cc}f_xf_h + u_cf_{xh}) = \tilde{f}_{gy} [g_x\varphi + g_h(u_{cc}f_xf_h + u_cf_{xh})].$$

Using the formulas for f_x , f_h , f_{xh} , and f_{hh} ,

$$\begin{aligned} g_x\varphi + g_h(u_{cc}f_xf_h + u_cf_{xh}) &= g_x \left[-u_{cc}\tilde{f}_g^2g_h^2 - u_c \left(\tilde{f}_{gg}g_h^2 + \tilde{f}_gg_{hh} \right) - u_{hh} \right] \\ &\quad + g_h \left[u_{cc}\tilde{f}_g^2g_xg_h + u_c \left(\tilde{f}_{gg}g_xg_h + \tilde{f}_gg_{xh} \right) \right] \\ &= -g_xu_{hh} + u_c\tilde{f}_g (g_hg_{xh} - g_xg_{hh}). \end{aligned}$$

Therefore,

$$f_{xy}\varphi + f_{hy}(u_{cc}f_xf_h + u_cf_{xh}) = \tilde{f}_{gy} [-g_xu_{hh} + u_c\tilde{f}_g (g_hg_{xh} - g_xg_{hh})].$$

If $u_c > 0$, $\tilde{f}_g > 0$, $\tilde{f}_{gy} > 0$, $g_x > 0$, and $u_{hh} < 0$, a sufficient condition for Condition (8) to hold is

$$g_hg_{xh} - g_xg_{hh} > 0,$$

that is,

$$g_{hh} < \frac{g_h}{g_x}g_{xh}.$$

For the hours condition, Proposition (3) becomes (recalling that for simplicity we assumed $u_{ch} = 0$)

$$u_{cc}f_xf_h + u_cf_{xh} > 0.$$

Substituting the same derivatives yields

$$u_{cc}\tilde{f}_g^2g_xg_h + u_c\tilde{f}_{gg}g_xg_h + u_c\tilde{f}_gg_{xh} > 0.$$

Dividing by $u_c\tilde{f}_gg_xg_h > 0$, we obtain

$$\frac{g_{xh}}{g_xg_h} > \tilde{f}_g \left(-\frac{u_{cc}}{u_c} \right) - \frac{\tilde{f}_{gg}}{\tilde{f}_g}.$$

This is the condition for hours to be increasing in skills in the bundle case. If $\tilde{f}_{gg} \leq 0$, then the right-hand side is positive, which implies that $g_{xh} > 0$ is necessary for $h_x > 0$.

B.3.3 Multiplicatively separable hours.

Suppose production is

$$f(x, y, h) = \bar{f}(x, y)z(y, h),$$

and preferences are separable CRRA,

$$u(c, h) = \frac{c^{1-\sigma}}{1-\sigma} - u(h), \quad c = f(x, y, h) - V(y),$$

with $V(y) \geq 0$. The relevant derivatives are $f_x = \bar{f}_x z$, $f_h = \bar{f} z_h$, $f_{xh} = \bar{f}_x z_h$, $f_{xy} = \bar{f}_{xy} z + \bar{f}_x z_y$, $f_{hy} = \bar{f}_y z_h + \bar{f} z_{hy}$, $f_{hh} = \bar{f} z_{hh}$. By Corollary (1), $h_x > 0$ if and only if

$$\sigma < \frac{f_{xh}(f - V)}{f_x f_h}.$$

Under multiplicative separability,

$$\frac{f_{xh}(f - V)}{f_x f_h} = \frac{\bar{f}_x z_h (f - V)}{(\bar{f}_x z)(\bar{f} z_h)} = \frac{f - V}{f} \leq 1,$$

where the last inequality uses $V(y) \geq 0$. Hence, $h_x > 0$ is impossible for any $\sigma \geq 1$. For sorting, Proposition (1) implies that PAM obtains if and only if

$$f_{xy} + f_{hy} h_x > 0.$$

Substituting the derivatives above gives⁸⁶

$$\bar{f}_{xy} z + \bar{f}_x z_y + (\bar{f}_y z_h + \bar{f} z_{hy}) h_x > 0.$$

B.4 Mechanism Details

Linking Global and Local Returns to hours The model mechanism sheds light on the link between the nature of hours in production, summarized by the complementarity parameter γ , and the convexity of earnings in hours. This link operates through sorting. To see this, recall that

⁸⁶This is the PAM condition stated in the text. In the more restrictive subcase $z(y, h) = z(h)$, it simplifies to

$$\bar{f}_{xy} z + \bar{f}_y z_h h_x > 0.$$

$$f(x, y, h) = A \left[\beta C(x, y)^\gamma + (1 - \beta) h^\gamma \right]^{1/\gamma}, \quad C(x, y) = (\alpha x^\rho + (1 - \alpha) y^\rho)^{1/\rho}.$$

Define

$$S(x, y, h) \equiv \beta C(x, y)^\gamma + (1 - \beta) h^\gamma.$$

Then the marginal product of hours is

$$f_h(x, y, h) = A(1 - \beta) h^{\gamma-1} S^{1/\gamma-1}.$$

Differentiate with respect to the skill-occupation bundle C:

$$f_{hC} = A\beta(1 - \beta)(1 - \gamma) h^{\gamma-1} C^{\gamma-1} S^{1/\gamma-2}.$$

So if $\gamma < 1$, and in particular if $\gamma < 0$,

$$f_{hC} > 0.$$

This is the first key point: better occupations, or better skill-job matches, have a higher marginal product of hours. Now, consider the role of sorting. Let equilibrium assignment be $y = \mu(x)$, and assume hours are increasing in skill, so that higher-hour workers are also higher-skill workers in better occupations. Then along the equilibrium allocation $x = \eta(h), y = \mu(\eta(h))$ and therefore

$$C(h) \equiv C(\eta(h), \mu(\eta(h))).$$

With positive assortative matching and $h_x^* > 0$, $C'(h) > 0$. In fact, since

$$C(h) = C(\eta(h), \mu(\eta(h))),$$

by the chain rule,

$$C'(h) = C_x(\eta(h), \mu(\eta(h))) \eta'(h) + C_y(\eta(h), \mu(\eta(h))) \mu'(\eta(h)) \eta'(h).$$

Using $\eta'(h) = 1/h_x(\eta(h))$, this becomes

$$C'(h) = \frac{C_x + C_y \mu'(\eta(h))}{h_x(\eta(h))}.$$

Since f_h depends on (x, y) only through $C(x, y)$, $f_{xh} = f_h C_x$, $f_{yh} = f_h C_y$, so equivalently

$$C'(h) = \frac{f_{xh} + f_{yh}\mu'(\eta(h))}{f_h C_x(\eta(h))}.$$

Under positive assortative matching, $\mu'(x) > 0$, and if hours are increasing in skill, $h_x(x) > 0$; since $C_x, C_y > 0$, this implies $C'(h) > 0$. Define the equilibrium marginal return to hours $m(h) \equiv f_h(C(h), h)$, it follows that $m'(h) = f_{hh} + f_{hC}C'(h)$. Because $f_{hC} > 0$ and $C'(h) > 0$, sorting contributes positively to $m'(h)$.

C Estimation Appendix

C.1 Estimation Details

Algorithm We minimize the SMM objective with a two-stage global–local procedure, separately for each year. In the global stage, we evaluate the objective at $N_1 = 2 \times 10^5$ low-discrepancy Sobol points spread quasi-uniformly over the parameter space. In the local stage, the $N^* = 10$ Sobol points with the lowest objective are used as starting values for a derivative-free Nelder–Mead simplex search, and we retain the parameter vector attaining the lowest objective across these restarts.

Weighting matrix and moment covariance The sampling uncertainty of the moments is estimated by bootstrap. Denote this full bootstrap covariance matrix by $\hat{C}_t = \widehat{\text{Var}}(\bar{m}_t)$. The weighting matrix used in estimation is diagonal and uses the bootstrap variances of the targeted moments: $\Omega_{t,jj} = \frac{1}{\widehat{\text{Var}}(\bar{m}_{jt})}$

SMM diagnostic checks For each year, I compute the numerical Jacobian of the vector of targeted model moments with respect to the estimated parameters,

$$J_t = \frac{\partial m_t(\theta)}{\partial \theta'}.$$

The relevant object for local identification is the weighted Jacobian $\sqrt{\Omega_t}J_t$. The rank tests indicate that $\sqrt{\Omega_t}J_t$ has full column rank in every year⁸⁷, so the six estimated parameters satisfy the local rank condition given the seven targeted moments, as shown in Tables C.1 and C.2.⁸⁸

⁸⁷I report only selected years for brevity, but the results are analogous for every year used in the estimation.

⁸⁸Because the productivity scale parameter A is measured in levels, I also report the same diagnostic after rescaling the A -column of the Jacobian to correspond to derivatives with respect to $\log A$. This rescaling does not change the model, but it makes the numerical conditioning of the local rank test easier to interpret.

Table C.1: Local rank condition, year $t = 1980$

Jacobian	Moments	Parameters	Rank	Min. singular value
$\sqrt{\Omega}J$	7	6	6	0.00224
$\sqrt{\Omega}J_{\log A}$	7	6	6	0.31402

Notes: The table reports the numerical rank of the weighted Jacobian of targeted moments with respect to estimated parameters. The second row rescales the A column to correspond to derivatives with respect to $\log A$.

Table C.2: Local rank condition, year $t = 2010$

Jacobian	Moments	Parameters	Rank	Min. singular value
$\sqrt{\Omega}J$	7	6	6	0.00166
$\sqrt{\Omega}J_{\log A}$	7	6	6	4.11713

Notes: The table reports the numerical rank of the weighted Jacobian of targeted moments with respect to estimated parameters. The second row rescales the A column to correspond to derivatives with respect to $\log A$.

I also report [Andrews, Gentzkow and Shapiro \(2017\)](#) sensitivity measures. Given the moment condition

$$g_t(\theta) = m_t(\theta) - \bar{m}_t,$$

the local sensitivity of the estimated parameters to perturbations in the data moments is computed in a reparameterized space used for the standard errors (see below), to deal with the fact that because the estimated A_t is very large, $J_t'\Omega_t J_t$ is mechanically poorly conditioned, as shown above. This is then mapped back to the original parameter units. Let

$$z_t = (\log A_t, \alpha_t, \beta_t, \gamma_t, \rho_t, b_t^y), \quad G_t = \frac{\partial \theta_t}{\partial z_t'} = \text{diag}(A_t, 1, 1, 1, 1, 1),$$

and let $J_t^z = J_t G_t$. The sensitivity matrix reported in the tables is⁸⁹

$$\Lambda_t = G_t (J_t^{z'} \Omega_t J_t^z)^{-1} J_t^{z'} \Omega_t. \quad (\text{C.1})$$

To make sensitivities comparable across moments measured in different units, I focus on standardized sensitivities, which scale each column by the bootstrap standard deviation of the corresponding data moment. The results show that the most influential

⁸⁹When the raw-parameter matrix is nonsingular, this is algebraically equivalent to $(J_t'\Omega_t J_t)^{-1} J_t'\Omega_t$; the reparameterization only improves numerical conditioning.

moments are wage dispersion and wage-hours moments rather than level moments. In particular, $\text{sd}(\log w)$, w_{90}/w_{10} , and w_{90}/w_{50} are central for the distributional and curvature parameters, while the hours-by-wage moments are especially important for γ , ρ , and b_y . This indicates that the joint wage-hours profile helps discipline the technology and assignment parameters. Results for selected representative years are reported in Tables C.3 and C.4.

Table C.3: Top-3 sensitivity moments by parameter, year $t = 1980$.

Parameter	Top 1	Top 2	Top 3
A	$\text{std}(w)$	w_{90}/w_{50}	$\mathbb{E}[h \mid d^w = 2]$
α	w_{90}/w_{10}	w_{90}/w_{50}	$\mathbb{E}[h \mid d^w = 5]$
β	w_{90}/w_{10}	w_{90}/w_{50}	$\mathbb{E}[h \mid d^w = 5]$
γ	$\mathbb{E}[h \mid d^w = 9]$	$\text{std}(w)$	w_{90}/w_{10}
ρ	w_{90}/w_{10}	$\mathbb{E}[h \mid d^w = 5]$	$\text{std}(w)$
b_y	w_{90}/w_{10}	$\mathbb{E}[h \mid d^w = 5]$	$\text{std}(w)$

Notes: The table reports, for each parameter, the three moments with the largest absolute standardized sensitivity in year 1980, computed using C.1.

Table C.4: Top-3 sensitivity moments by parameter, year $t = 2010$.

Parameter	Top 1	Top 2	Top 3
A	w_{90}/w_{50}	$\text{std}(w)$	$\mathbb{E}[h \mid d^w = 5]$
α	$\text{std}(w)$	$\mathbb{E}[h \mid d^w = 5]$	w_{90}/w_{50}
β	w_{90}/w_{50}	$\mathbb{E}[h \mid d^w = 5]$	$\text{std}(w)$
γ	$\mathbb{E}[h \mid d^w = 9]$	$\mathbb{E}[h \mid d^w = 5]$	w_{90}/w_{10}
ρ	$\mathbb{E}[h \mid d^w = 5]$	w_{90}/w_{50}	$\mathbb{E}[h \mid d^w = 9]$
b_y	$\mathbb{E}[h \mid d^w = 5]$	$\mathbb{E}[h \mid d^w = 2]$	$\mathbb{E}[h \mid d^w = 9]$

Notes: The table reports, for each parameter, the three moments with the largest absolute standardized sensitivity in year 2010, computed using C.1.

Standard Errors SMM standard errors are computed using

$$\widehat{\text{Var}}(\hat{\theta}_t) = (J'_t \Omega_t J_t)^{-1} J'_t \Omega_t \hat{C}_t \Omega_t J_t (J'_t \Omega_t J_t)^{-1},$$

where $\hat{C}_t$ is the full bootstrap covariance matrix of the data moments. The weighting matrix Ω_t is the diagonal inverse-variance matrix used in estimation, constructed from the

diagonal of $\widehat{C}_t$. Thus, Ω_t uses only the bootstrap variances, while the sandwich formula uses the full covariance matrix of the data moments.⁹⁰

C.2 Additional Model Validation

We now show a series of model-data comparison exercises for a few key untargeted moments. Figure C.1 plots average hours worked by wage decile for two selected years. Overall, the Figure shows that the model tracks data closely, with the exception of the 10th decile in 2018, for which the model underpredicts hours worked.

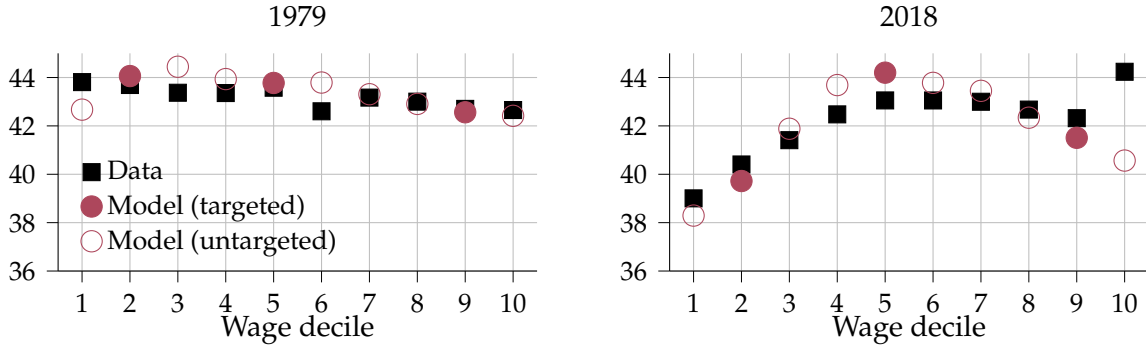


Figure C.1: Hours worked by wage decile: model vs. data.

Notes: The figure compares average hours worked by wage decile in the model and in the data for 1979 and 2018. Black squares denote the data. Red filled circles denote the model moments that are directly targeted in the estimation (wage deciles 2, 5, and 9). Red open circles denote untargeted model implications for the remaining wage deciles.

Figure C.2 compares the probability of working more than 50 hours per week across wage deciles in the model and in the data. Figure C.3 compares the weekly-earnings distributions. For both Figures, the model produces a decent fit of these untargeted moments; in particular, it correctly captures the declining and increasing probability of working long hours (in 1979 and 2018, respectively) and closely tracks the weekly earnings distribution, with the main miss being the upper part of the distribution in 2018.

⁹⁰Because A_t enters in levels and is large, $J_t' \Omega_t J_t$ is poorly conditioned in the raw parameters (the same scaling issue visible in the rank diagnostics above). We therefore use the reparameterized vector $z_t = (\log A_t, \alpha_t, \beta_t, \gamma_t, \rho_t, b_t^y)$, i.e. with J_t replaced by $J_t^z = J_t G_t$, where $G_t = \partial \theta_t / \partial z_t = \text{diag}(A_t, 1, 1, 1, 1, 1)$ is evaluated at the estimate and rescales only the A -column. In z -space the matrix $J_t^{z'} \Omega_t J_t^z$ is well-conditioned and I recover the raw-parameter covariance by the delta method,

$$\widehat{\text{Var}}(\hat{\theta}_t) = G_t \widehat{\text{Var}}(\hat{z}_t) G_t', \quad \widehat{\text{Var}}(\hat{z}_t) = (J_t^{z'} \Omega_t J_t^z)^{-1} J_t^{z'} \Omega_t \widehat{C}_t \Omega_t J_t^z (J_t^{z'} \Omega_t J_t^z)^{-1}.$$

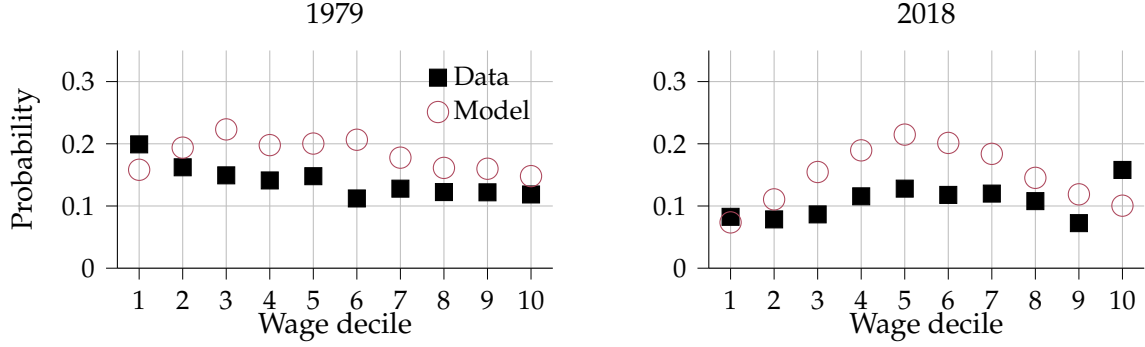


Figure C.2: Probability of working long hours by wage decile: model vs. data.

Notes: The figure compares the probability of working more than 50 hours per week by wage decile in the model and in the data for 1979 and 2018. Wage deciles are computed separately within each year. Black squares denote the CPS-ORG data and red open circles denote the model.

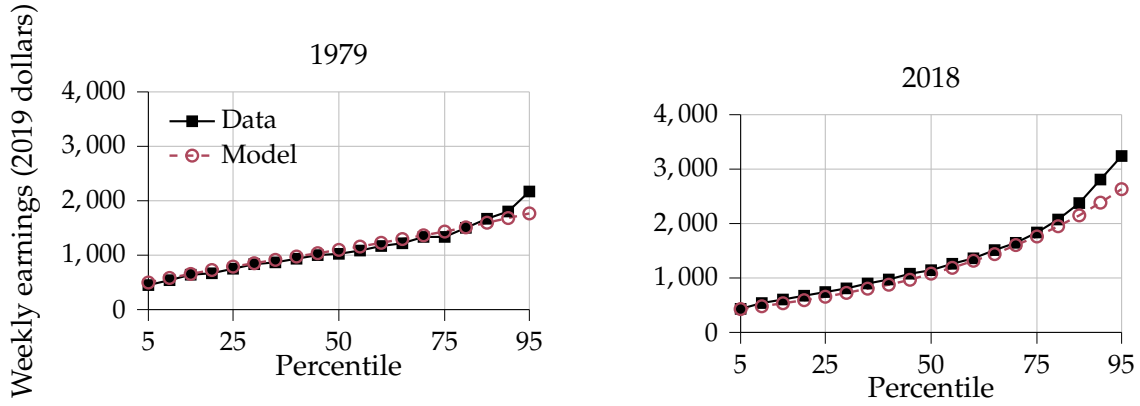


Figure C.3: Weekly-earnings distribution: model vs. data.

Notes: The figure plots weighted weekly-earnings quantiles in the model and in the data for 1979 and 2018, at five-percentage-point intervals from the 5th through the 95th percentile. Data earnings are CEPR ‘weekpay’ converted to real 2019 dollars using the year-specific price level. Model earnings are the weekly earnings generated by the estimated model.

C.3 Additional Counterfactuals

Interaction of HBTC and sorting: decomposition Figure C.4 contains the decomposition of income inequality in Figure 12 in each of its three channels.

D Implications Appendix

D.1 Stochastic Extension

This subsection presents the stochastic extension used in the counterfactual exercises. The extension follows the distance-to-frontier approach (Galichon *et al.*, 2019) to matching with imperfectly transferable utility (ITU). The key additional ingredient is that a contract

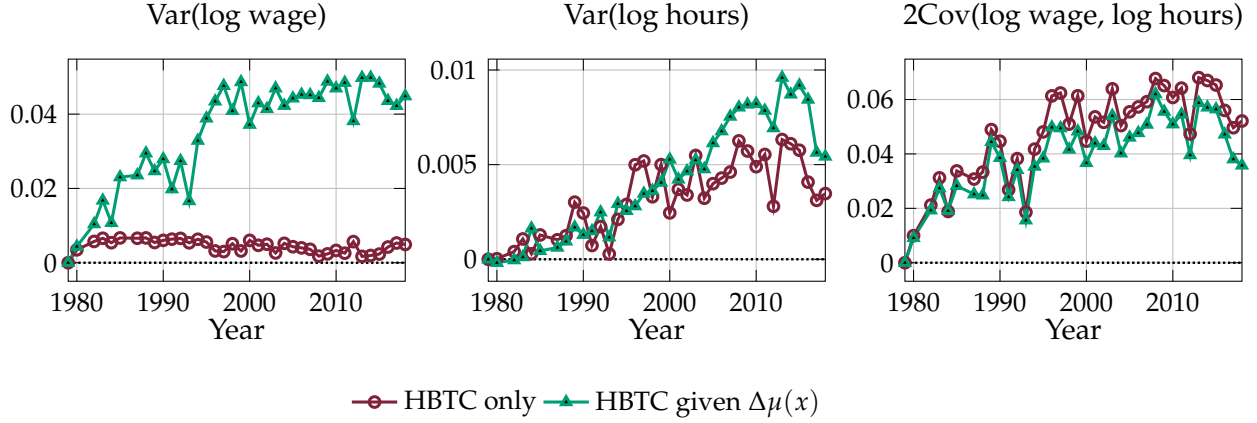


Figure C.4: Marginal effects: decomposition.

Notes: The figure displays the changes in the three components of the variance of log income in the model economy when estimated changes in γ are fed into the 1979 estimated economy, leaving all other parameters unchanged, relative to the level of the variance of log income in 1979 ("HBTC only"); the change in the same components when both estimated changes in γ and in ρ, b^y are fed into the 1979 estimated economy relative to when only ρ, b^y are changing ("HBTC given $\Delta\mu$ ").

is now a pair (y, h) , where y is the (discrete) job type and $h \in \mathcal{H}$ is hours worked, also discrete.

Hours wedge. We study a reduced-form wedge that raises the nonpecuniary cost of supplying hours:⁹¹

$$\psi_\tau(h) = (1 + \tau)\psi(h), \quad \tau \geq 0. \quad (\text{D.1})$$

Timing and shocks. The stochastic model is a simultaneous contract-choice model, i.e. hours are part of the contract. For each worker type x , the choice set is

$$\{(y, h) : y \in Y, h \in \mathcal{H}\} \cup \{\emptyset\},$$

where $\emptyset$ denotes remaining unmatched. Likewise, for each job type y , the choice set is

$$\{(x, h) : x \in X, h \in \mathcal{H}\} \cup \{\emptyset\}.$$

Each worker of type x receives i.i.d. type-I extreme-value shocks

$$\{\varepsilon_{xyh}\}_{y \in Y, h \in \mathcal{H}} \quad \text{and} \quad \varepsilon_{x0}$$

⁹¹Here, with a slight abuse of notation, $\psi(h) = \psi h^{1+1/\theta} / (1 + 1/\theta)$ denotes the full hours-cost function. Multiplying it by $1 + \tau$ is equivalent to scaling the coefficient ψ as in the main text. When $\tau = 0$, the economy coincides with the baseline. When $\tau > 0$, long hours become more costly in utility terms.

over the contract alternatives and the outside option. Each job of type y receives i.i.d. type-I extreme-value shocks

$$\{\eta_{xyh}\}_{x \in X, h \in \mathcal{H}} \quad \text{and} \quad \eta_{0y}.$$

All shocks are independent across sides and alternatives and have common scale parameter $\sigma_\delta > 0$.⁹²

Bargaining sets. Fix a worker type x , a job type y , and an hours contract $h \in \mathcal{H}$. If the worker consumes $c \geq 0$, then worker utility is $u(c) - \psi_\tau(h)$ and the job receives the residual $f(x, y, h) - c$. The corresponding contract-level bargaining set is

$$F^\tau(x, y, h) = \left\{ (\tilde{u}, \tilde{v}) \in \mathbb{R}^2 : \exists c \geq 0 \text{ s.t. } \tilde{u} \leq u(c) - \psi_\tau(h), \quad \tilde{v} \leq f(x, y, h) - c \right\}. \quad (\text{D.2})$$

Distance-to-frontier. For any $(U, V) \in \mathbb{R}^2$, define the contract-level distance-to-frontier as

$$D_{xyh}^\tau(U, V) = \inf \{ z \in \mathbb{R} : (U - z, V - z) \in F^\tau(x, y, h) \}. \quad (\text{D.3})$$

Using (D.2), this distance can be written equivalently as

$$D_{xyh}^\tau(U, V) = \min_{c \geq 0} \max \{ U - (u(c) - \psi_\tau(h)), V - (f(x, y, h) - c) \}. \quad (\text{D.4})$$

Equation (D.4) is the contract-level analogue of the distance representation in [Galichon et al. \(2019\)](#).

Systematic utilities. Let U_{xyh} denote the worker-side systematic utility associated with contract (x, y, h) and let V_{xyh} denote the corresponding job-side systematic payoff. Under the EV1 specification, the standard log-odds formulas imply

$$U_{xyh} = u_0 + \sigma_\delta \log \left(\frac{\mu(x, y, h)}{\mu(x, \emptyset)} \right), \quad (\text{D.5})$$

and

$$V_{xyh} = v_0 + \sigma_\delta \log \left(\frac{\mu(x, y, h)}{\mu(\emptyset, y)} \right). \quad (\text{D.6})$$

Define the type-level potentials $U(x) = u_0 - \sigma_\delta \log \mu(x, \emptyset)$, $V(y) = v_0 - \sigma_\delta \log \mu(\emptyset, y)$. Then (D.5) and (D.6) can be rewritten as

$$U_{xyh} = U(x) + \sigma_\delta \log \mu(x, y, h), \quad V_{xyh} = V(y) + \sigma_\delta \log \mu(x, y, h). \quad (\text{D.7})$$

⁹²Thus, a single scale parameter governs both the dispersion of matching across partner types and the dispersion of hours choices within matches.

At equilibrium, it must be that

$$D_{xyh}^\tau(U_{xyh}, V_{xyh}) = 0. \quad (\text{D.8})$$

Using the translation property of the distance-to-frontier function⁹³ and substituting (D.7) into (D.8), we obtain

$$0 = D_{xyh}^\tau(U(x), V(y)) + \sigma_\delta \log \mu(x, y, h).$$

Hence the equilibrium mass of contracts (x, y, h) is

$$\mu(x, y, h) = \exp \left(-\frac{D_{xyh}^\tau(U(x), V(y))}{\sigma_\delta} \right). \quad (\text{D.9})$$

Pair masses. The mass of matches of type (x, y) is the sum of contract masses across hours:

$$\mu(x, y) = \sum_{h \in \mathcal{H}} \mu(x, y, h). \quad (\text{D.10})$$

Substituting (D.9) into (D.10) yields

$$\mu(x, y) = \sum_{h \in \mathcal{H}} \exp \left(-\frac{D_{xyh}^\tau(U(x), V(y))}{\sigma_\delta} \right).$$

Define the pair-level distance

$$D_{xy}^\tau(U, V) = -\sigma_\delta \log \sum_{h \in \mathcal{H}} \exp \left(-\frac{D_{xyh}^\tau(U, V)}{\sigma_\delta} \right). \quad (\text{D.11})$$

Then the pair mass can be written compactly as

$$\mu(x, y) = \exp \left(-\frac{D_{xy}^\tau(U(x), V(y))}{\sigma_\delta} \right). \quad (\text{D.12})$$

Matching equations. The unmatched masses are recovered from the type-level potentials:

$$\mu(x, \emptyset) = \exp \left(\frac{u_0 - U(x)}{\sigma_\delta} \right), \quad \mu(\emptyset, y) = \exp \left(\frac{v_0 - V(y)}{\sigma_\delta} \right). \quad (\text{D.13})$$

⁹³Explicitly,

$$D_{xyh}^\tau(a + t, b + t) = D_{xyh}^\tau(a, b) + t,$$

Combining (D.12) with (D.13), equilibrium potentials solve

$$\exp\left(\frac{u_0 - U(x)}{\sigma_\delta}\right) + \sum_{y \in Y} \exp\left(-\frac{D_{xy}^\tau(U(x), V(y))}{\sigma_\delta}\right) = n_x, \quad \forall x \in X, \quad (\text{D.14})$$

and

$$\exp\left(\frac{v_0 - V(y)}{\sigma_\delta}\right) + \sum_{x \in X} \exp\left(-\frac{D_{xy}^\tau(U(x), V(y))}{\sigma_\delta}\right) = m_y, \quad \forall y \in Y. \quad (\text{D.15})$$

Hours. Conditional on the pair (x, y) being formed, the probability of choosing hours h is

$$\pi^\tau(h \mid x, y) = \frac{\exp\left(-\frac{D_{xyh}^\tau(U(x), V(y))}{\sigma_\delta}\right)}{\sum_{k \in \mathcal{H}} \exp\left(-\frac{D_{xyk}^\tau(U(x), V(y))}{\sigma_\delta}\right)}. \quad (\text{D.16})$$

Hence the mass of contracts (x, y, h) is

$$\mu^\tau(x, y, h) = \mu(x, y) \pi^\tau(h \mid x, y). \quad (\text{D.17})$$

Solution method. Under logit heterogeneity, the stochastic extension can be written as a matching-function equilibrium in the sense of Galichon *et al.* (2019). In particular, the theory above defines a matching function

$$M_{xy}^\tau(\mu(x, \emptyset), \mu(\emptyset, y)) = \exp\left(-\frac{D_{xy}^\tau(U(x), V(y))}{\sigma_\delta}\right),$$

where $U(x) = u_0 - \sigma_\delta \log \mu(x, \emptyset)$ and $V(y) = v_0 - \sigma_\delta \log \mu(\emptyset, y)$. Hence equilibrium reduces to solving the nonlinear system (D.14)–(D.15) in the unmatched masses.⁹⁴

Results Figure D.1 illustrates the effects of a change in the hours wedge τ by comparing outcomes for bottom- and top-skill workers under the HBTC benchmark and under the no-HBTC counterfactual.⁹⁵ Under HBTC, the wedge reduces the probability that

⁹⁴Following Galichon *et al.* (2019), we initialize one side's unmatched masses at a sufficiently high value and iterate between row and column updates: holding $\mu(\emptyset, y)$ fixed, we solve each worker equation for $\mu(x, \emptyset)$; holding the resulting $\mu(x, \emptyset)$ fixed, we solve each job equation for $\mu(\emptyset, y)$. This is an application of the iterative proportional fitting procedure (IPFP) adapted to the ITU-logit matching-function system. Under the same full-support and regularity assumptions as in Galichon *et al.* (2019), the induced matching function inherits the properties needed for monotone convergence, and the resulting equilibrium is unique. This is the same logic underlying the computational procedure in Bandiera *et al.* (2024).

⁹⁵The paper's baseline counterfactuals take the terminal-year parameter vector as the benchmark technology. The no-HBTC comparison replaces only the hours-complementarity parameter γ with its 1979

top workers occupy top jobs and raises the corresponding probability for bottom workers (middle panel). Hence, the distortion worsens the allocation of talent exactly in the region where long hours are most productive. Without HBTC, by contrast, the reallocation pattern can even go in the opposite direction. In other words, whether the same hours wedge worsens or attenuates mismatch depends on the production technology governing the value of hours in top matches. Moreover, under HBTC, the same wedge reduces income proportionally more at the top than at the bottom; the difference under the no-HBTC case is much smaller (left panel). This occurs despite the reduction in long hours occurring quite uniformly across skills (right panel). Depending on whether hours are particularly complementary in top jobs, workers in those jobs are those who suffer the largest consequences: absent HBTC, long hours can partially act as a substitute margin, allowing lower-skill workers to compensate for lower skill by supplying more hours. A wedge on hours then weakens that compensation margin. Under HBTC, however, long hours are part of the comparative advantage of high-skill workers in high-quality jobs. Raising the cost of hours therefore reduces the value of precisely those top contracts, pushes some high-skill workers away from them, and allows lower-skill workers to move in.

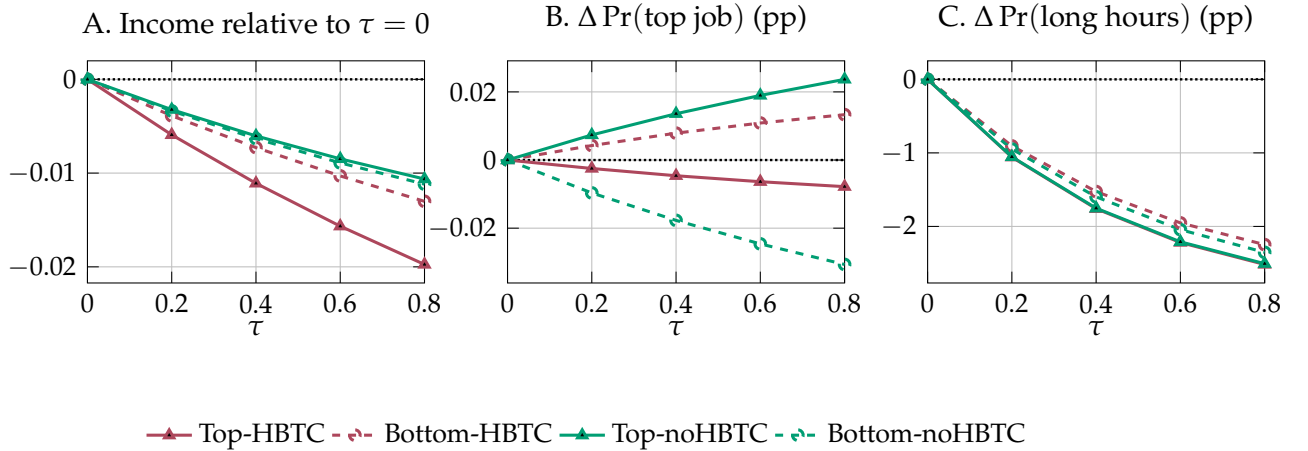


Figure D.1: Top versus bottom workers: effects of τ .

Notes: The figure compares bottom- and top-skill workers under the HBTC benchmark and under the no-HBTC counterfactual. It plots income relative to $\tau = 0$, the change in the unconditional probability of obtaining a top job, and the change in the unconditional probability of working long hours. The HBTC benchmark is defined as the simulated model using the estimated value of γ in 2018. The no-HBTC counterfactual is defined as the simulated model using the estimated value of γ in 1979. The remaining parameter values are held at their 2018 estimated values. The shocks dispersion parameter σ_δ is fixed at 0.5.

estimate while keeping all other parameters at their terminal-year values.

D.2 Elasticities

We employ elasticity concepts borrowed from [Scheuer and Werning \(2017\)](#). In particular, we use fixed-assignment elasticities (how much earnings respond if the worker changes hours but stays at the current match/job) and full-assignment elasticities (how much earnings respond if the worker changes hours and is also allowed to move along the equilibrium assignment/hedonic schedule).⁹⁶ This appendix compares these elasticities in the estimated economy.

Baseline equilibrium and local schedule representation Fix a baseline equilibrium under tax schedule T . For each worker type x in the region of interest, let $y_0(x)$ be the baseline job, $h_0(x)$ be baseline hours, and $w_0(x)$ baseline earnings. For a fixed worker type x , define *fixed-assignment* schedule: holding the baseline job $y_0(x)$ fixed,⁹⁷

$$\widehat{W}_x(h) \equiv f(x, y_0(x), h) - V(y_0(x)).$$

Moreover, define the *full* schedule: allowing the worker to move along the baseline job menu,

$$W_x(h) \equiv \max_y \{f(x, y, h) - V(y)\}.$$

Let $y_x(h)$ denote the argmax in the definition of $W_x(h)$. At the baseline point (y_0, h_0, w_0) ,

$$W_x(h_0) = \widehat{W}_x(h_0) = w_0.$$

The two schedules are tangent at h_0 by the envelope theorem:

$$W'_x(h_0) = \widehat{W}'_x(h_0).$$

Quantitative relevance We now quantify how much larger the required full-assignment elasticity is than the fixed-assignment elasticity in the estimated economy, and how this correction changes when the hours-biased technology channel is shut down. Let $\varepsilon_c(x)$ and $\widehat{\varepsilon}_c(x)$ denote the compensated earnings elasticities with respect to a local change in

⁹⁶In the spirit of [Scheuer and Werning \(2017\)](#), the elasticities computed below are partial-equilibrium elasticities, obtained by perturbing the worker's local tax wedge while holding the relevant baseline earnings schedules fixed. This is sufficient for the mechanism discussed in the main text, but it's possible to analyze how sorting itself is impacted by progressive taxation. This is beyond the scope of this exercise, and results on the topic can be found in [Vereshchagina \(2021\)](#).

⁹⁷Recall that earnings(income) are defined as

$$w(x, y, h) = f(x, y, h) - V(y),$$

where $V(y)$ is the equilibrium value associated with job y , or job/firm profits.

the net-of-tax rate under full and fixed assignment, respectively. For each year, I compute the top-tail average adjustment

$$\overline{\mathcal{R}}_{t,p} \equiv \mathbb{E}_t \left[\frac{\varepsilon_c(x)}{\widehat{\varepsilon}_c(x)} \middle| x \in \text{top } p\% \text{ of the earnings distribution} \right],$$

with $p = 1$ in the baseline specification, and compare it to the corresponding object in a no-HBTC counterfactual.⁹⁸ This exercise isolates how much HBTC changes the wedge between the required elasticity and the within-job one in the upper tail of the earnings distribution.

Figure D.2 shows that the average ratio of the full-assignment to the fixed-assignment compensated earnings elasticity in the top 1% exceeds one in both economies throughout the sample, so that the assignment margin matters quantitatively in the upper tail. At the same time, the top-tail adjustment is systematically larger in the estimated economy than in the no-HBTC counterfactual, and particularly so in recent decades.

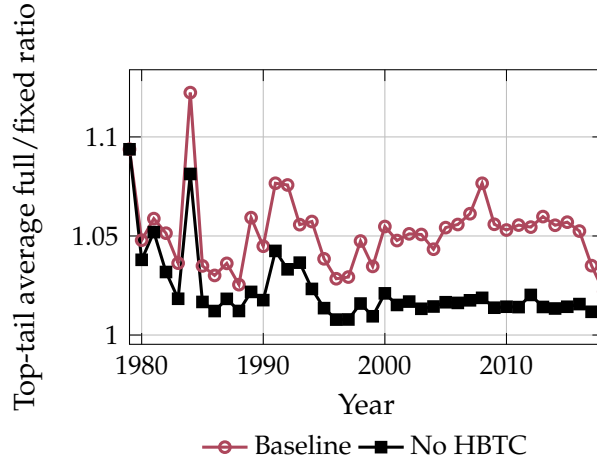


Figure D.2: Top-tail elasticity adjustment over time.

Notes: The figure plots the top-tail average ratio of the full compensated elasticity to the fixed-assignment compensated elasticity over time in the baseline model and in the no-HBTC counterfactual.

D.3 Additional Supporting Figures

To explore how the rise in hours worked at the top is due to life-cycle motives, I divide workers in the top wage decile into four age groups. The top wage decile is defined separately in each year using the full baseline sample. Figure D.3 reports the probability of working more than 50 hours per week in levels and as changes relative to 1979. Between

⁹⁸The no-HBTC economy is constructed exactly as in the main text: I hold fixed γ at its initial level and re-solve the equilibrium in each year.

1979 and 2018, the probability rises and then declines among workers aged 25–34, while it rises among both workers aged 45 and above. Figure D.4 presents the analogous exercise for average weekly hours, showing similar age patterns.

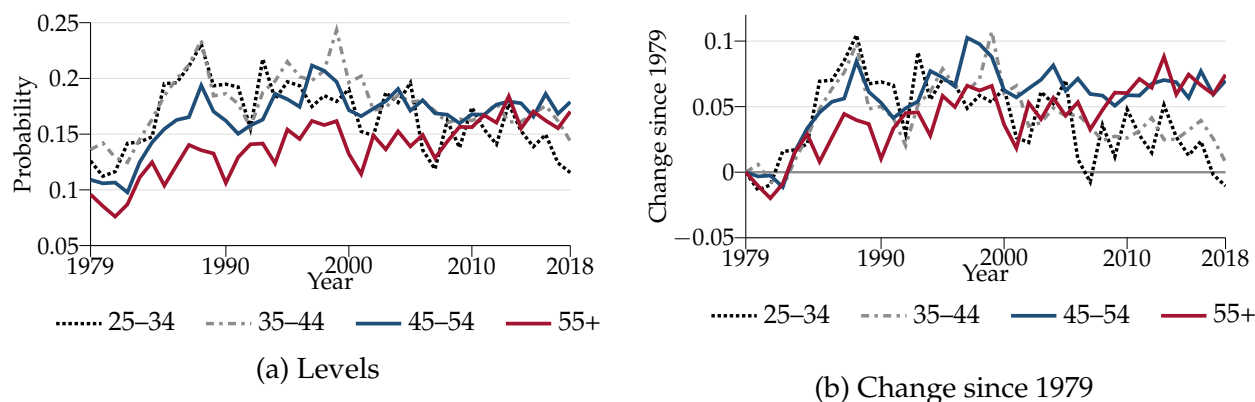


Figure D.3: Long Hours in the Top Wage Decile, by Age Group.

Notes: Panel (a) reports the weighted probability of working more than 50 hours per week among workers in the top wage decile, separately by age group. Panel (b) reports the change in that probability relative to 1979. The top wage decile is recomputed in the full baseline sample in each year before workers are separated by age. The groups are ages 25–34, 35–44, 45–54, and 55–64. Source: CPS-ORG.

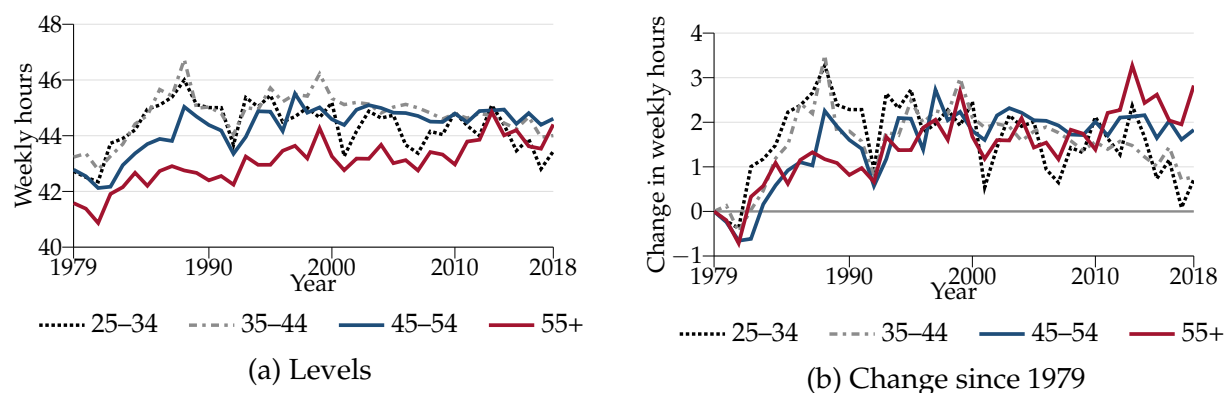


Figure D.4: Average Weekly Hours in the Top Wage Decile, by Age Group.

Notes: Panel (a) reports weighted average weekly hours among workers in the top wage decile, separately by age group. Panel (b) reports the change in average weekly hours relative to 1979, measured in hours. The top wage decile is recomputed in the full baseline sample in each year before workers are separated by age. The groups are ages 25–34, 35–44, 45–54, and 55–64. Source: CPS-ORG.